



**DESIGN AND IMPLEMENTATION OF A FREELANCE MARKET PLACE FOR
NILE STUDENTS (CAMPUS GIGS)**

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**A PROJECT SUBMITTED TO THE DEPARTMENT OF INFORMATION
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AWARD OF DEGREE OF BACHELOR OF SCIENCE IN INFORMATION
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NILE UNIVERSITY OF NIGERIA.

JUNE 2026

DECLARATION

I hereby declare that the work in this project is my own except for quotations and summaries which have been duly acknowledged.

Student :

Date :

Supervisor :

SUPERVISOR'S NAME

DEDICATION

I dedicate this project to Almighty God for His guidance, wisdom, strength, and protection throughout the course of this study. I also dedicate this work to our parents, guardians, and families for their continuous support, encouragement, sacrifices, and prayers during our academic journey. Furthermore, this project is dedicated to our lecturers and mentors whose knowledge, guidance, and motivation contributed immensely to the successful completion of this research work. Finally, we dedicate this project to all students of Nile University of Nigeria and aspiring young professionals who seek to use technology to create opportunities, foster collaboration, and drive innovation within academic communities.

ACKNOWLEDGEMENT

The completion of this project was a collective effort, and we wish to express our profound gratitude to those who guided us through the process. First and foremost, we thank the Almighty for the wisdom, strength, and health granted to us throughout the duration of this research and development phase. We extend our deepest gratitude to our project supervisor, whose expert guidance, critical feedback, and patience were instrumental in refining our system design and documentation. We also thank the faculty and staff of our department for providing a conducive environment for learning and innovation. The foundational knowledge we gained in our previous years of study provided the building blocks for creating the system. Lastly, we thank our peers who participated in our system analysis and user testing, providing the feedback necessary to refine the interfaces in the system. Your input helped us create a tool that truly addresses the needs of student freelancers and job posters.

ABSTRACT

Campus Gigs is a web-based freelance marketplace designed to connect students within a university environment by enabling them to offer and request freelance services efficiently. The platform provides a centralized system where student clients can post jobs, review applications, communicate with freelancers, and process payments, while student freelancers can browse opportunities, apply for jobs, submit completed work, and receive ratings and payments. The system aims to solve common challenges faced by students in finding reliable freelance opportunities and affordable campus-based services by creating a secure and organized digital marketplace. The platform also includes an administrator module for managing users, moderating activities, handling disputes, and monitoring transactions to ensure smooth platform operation. Campus Gigs was designed using modern software engineering principles and UML modeling techniques, including use case diagrams, activity diagrams, and deployment diagrams, to ensure proper system structure and scalability. The system architecture supports secure communication, user authentication, job management, messaging, notifications, and payment processing. The project demonstrates how technology can improve student collaboration, promote entrepreneurship, and create flexible earning opportunities within academic communities through an intuitive and accessible online platform.

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CHAPTER ONE

INTRODUCTION

In this chapter, we discuss the study's background, the problem, why it's essential, the goals, the scope, and how it will be set up. It explains why Campus Gigs, a new freelance marketplace for students, is vital and sets the stage for the study. Campus Gigs makes it simple, quick, and safe to locate freelance work on campus.

1.1 BACKGROUND OF THE STUDY

The growth of digital technologies and the gig economy has had a huge impact on higher education and professional growth. In the last few years, the word "work" has altered. People are not working long-term jobs anymore; instead, they are doing work that is flexible and project-focused. A number of college students have loved this trend [1]. Students today do not just sit back and learn; they also produce things. Many of them already have advanced skills in graphic design, software development, and specialised tutoring before they ever graduate. But there is a tremendous difference between letting Students apply these abilities in their own classrooms and possessing them.

This project lets Students meet new people and use what they have learnt in school to improve the local economy. The platform's fundamental idea is "Connect. Collaborate. Create." The site intends to link two primary groups of people: Student Clients (Job Posters), who require help with certain activities, and Student Freelancers (Job Applicants), who wish to share what they know with other students. The program connects professionals at universities so they may meet up and help one other.

The problems that both informal networking and global freelancing platforms are encountering right now illustrate how much we need a system like this. Students desire to work in the gig economy, but they often have huge issues, like not being able to prove their reputation and having to compete with more experienced workers on worldwide platforms [2]. The only way to use this project is with a secure Authentication Module and a Landing Page with a hero banner that displays youngsters working together. This makes the students' surroundings perfect. People can search for jobs by the name of the project or the type of work they are skilled at. This makes sure that the correct people can find jobs and apply for them.

The two important aspects to this background are professional growth and project management. When collaborating with other learners in a normal manner, students are not always aware of

what to do. This may lead to failure to do work and failure to meet expectations. This paper alleviates these problems through the use of potent Job Management Module and a Messaging Module that ease communication among themselves by chatting to each other, sharing files and getting real-time updates on the status of their jobs. The system makes users feel that they are having professional duty by letting them see their progress between the start to in progress and completed. The User Profile Module also allows students to create a digital portfolio which displays their talents, department, and a list of work they have done. This portfolio could be seen by their friends and their would-be employers.

This paper is based on the assumption that a properly structured, mobile device friendly platform can assist students in preparing to enter the job market in a significant number [3]. The platform provides students with practical experience, enables them to earn income and provides students with access to people that can guide them in their careers. Admin Module ensures that all is secure and equitable. The project will be a shift to a more collaborative and economically empowered campus, with the use of modern digital technology to find solutions to specific professional issues [4] [33] [34].

1.2 PROBLEM STATEMENT

Today schools educate students about how to code, design and teach, however, they do not instruct them on how to apply these skills in their respective societies [5]. Students are currently appealing using broken social networking sites such as WhatsApp, Telegram and physical notice boards to seek assistance in their projects. Such unofficial channels are inevitably chaotic and the advertisement of jobs might be easily lost in the discussions which do not refer to the advertisement. This implies that both Student Client (Job Poster) and Student Freelancer (Job Applicant) are deprived of opportunities.

Students also find it hard to start working with such global freelancing platforms as Upwork and Fiverr. The entry of students with little experience in these industries makes it difficult to secure the first job in these industries since there are very many experienced workers who know what they are doing [6]. It does not have a structure of a central Job Feed where the most obvious filters are Category, Deadline, and Level so that students spend more time searching for a job rather than working.

There is much more to be done in order to gain trust and carry out initiatives. During less formal interactions, one does not have a set method to verify the identity of a peer or their skills. The fact that Authentication Modules and User Profiles are not clearly structured meant that people did not have the idea of what to do and may even attempt to cheat. In the majority of cases, the partnership is not a successful experience due to the fact that the individuals involved fail to communicate with each other and walk away [7]. It does not have a separate Messaging Module to monitor the statuses of projects as they transition between "Start" and the "In Progress" to the "Completed" status. Lastly, the students might be unprofessional and post unfair content due to the lack of an Admin Module to monitor the institution [8]. We require an online platform to operate through mobile services directly in real-time and transform these engagements into professional and open market.

1.3 SIGNIFICANCE OF THE STUDY

The social, economic and professional requirements of the academic community are very important as this is a student-to-student platform to discuss. This plan renders the gig economy a permanent feature on campus. It enables the students, teachers and the school, in general, to utilize their ability in an organized manner.

This topic is crucial to the career of the students as it allows them to develop a work-related image that they can present. The User Profile Module of this system is not similar to other social media profiles; it is expected to be a digital portfolio. Completed Jobs History This is the history of the jobs that a student has completed on the site. This demonstrates that they are not only experienced in things such as coding, design, or tutoring [9]. Research indicates that, the undergraduate students that have exposure to projects are far more equipped to the job market since they can demonstrate to the future employers that they will be able to tackle practical issues in the real world. These portfolios are even more credible since the system is used to demonstrate and verify the level of skills with the help of badges. The service helps Students pay for their academic and living costs in a safe and authentic way. A local marketplace aids students by making it easier for them to locate short-term jobs that fit into their busy academic schedules. Students feel safe working together on money difficulties because the Authentication Module, which requires them to sign in with their school email address, makes the environment highly safe [11] [14]. This localized trust is much better than worldwide platforms, where clients might not feel safe if they do not have a link to an institution.

The study reveals how schools and universities are making role-based collaborative systems from both a technical and an administrative point of view. The Admin Dashboard is very significant since it gives the organisation information based on data. The "total jobs" and "most active category" figures show university authorities what talents students are most interested in. This might help them adjust the curriculum in the future [12]. The system's focus on a "clean and visual" card-based UI and mobile responsiveness also makes sure that today's students, who largely use their phones, can rapidly use and access the platform.

The study's major goal is to help make the school a place where students can "Connect. Collaborate. Create." It's not just a job board; it's also a location where people can seek help from the community, progress professionally, and become financially independent. [13] [17].

1.4 AIM AND OBJECTIVES

Setting clear goals is highly crucial throughout the whole life cycle of a software development project. This is especially true for projects that wish to develop a new area, like a student marketplace. This section talks about the basic purpose of the platform and the many technical and functional tasks that need to be done to make it work.

1.4.1 Aim

The aim of this project is to develop a Freelance marketplace (Campus Gigs) for the Nile students

1.4.2 Objectives

To achieve the aforementioned aim, the following specific objectives have been identified:

- i. To design an intuitive Freelance marketplace System.
- ii. To design a responsive and user-friendly interface that supports accessibility across desktop and mobile devices.
- iii. To comprehensively test the system to ensure it meets all functional and non-functional requirements, including performance, unit, and system.

1.5 SCOPE AND LIMITATION OF STUDY.

It is important to have stringent limits to a technology project to ensure that it remains on schedule and is completed within the time frame. This part discusses what the platform is capable and incapable of currently.

1.5.1 Scope

The key objective of this project is to establish and sustain a Freelance marketplace Nile students. The platform will be linked to Nile university students and clients to provide freelance work including job posting, messaging, and payment functionality to provide a better user experience. The design will serve the special needs of students in the freelance economy that have special schedule and financial limitations.

1.5.2 Limitations

Although the platform is rather comprehensive, there are a number of constraints to the current research:

- i. Vitality Margin Outlawing Large Financial Transactions.
- ii. Geographical and Institutional Constraint: The platform is not yet configured to scale to the cross-campus and global marketplace, and is only optimized to Nile university students in its present state.
- iii. Lack of high-tech AI-based job matching capabilities.
- iv. Technical Dependencies The real-time nature of the system (such as messaging) will rely on the consistent availability of internet connectivity and server-side resources; performance can be affected by high-latency.
- v. Mobile Experience: The web interface is completely responsive; however, at this point the study does not involve the creation of a native mobile app (iOS or Android)

1.6 ORGANISATION OF THE STUDY

The research organization has been provided in such a way that it takes a detailed step-by-step description of how the platform was developed and its theoretical basis. The chapters play a certain role in the software development lifecycle (SDLC):

Chapter One: Introduction - It is the roadmap to the whole project. It addresses the weaknesses of the existing student gig economy and why a single location should be considered for students

to seek employment. It discusses the many things that should be accomplished, the challenges that should be resolved, and the objectives that should be achieved.

Chapter Two: Literature Review - The chapter gives the academic and industry background of the study. It examines the history of the gig economy, the emergence of the niche market, and the technical architectures of modern web development. It also uses at minimum 40 peer-reviewed resources, research papers, and technical documentations to support the existence of the project in the modern digital environment.

Chapter Three: System Analysis and Design -This chapter shifts the focus on the theory to the technical planning. It consists of the detailed examination of the Waterfall methodology adopted in the project, the examination of current manual operations and the design requirements of the new system. UML diagrams, such as Use Case Diagrams, Activity Diagrams, and the System Architecture, are used to a great extent in this section, and they visualize the data flow between students and the database.

Chapter Four: System Implementation and Testing - Chapter four explains the very construction stage. It records the decision of programming languages and structures utilized in the development of the system [23].

Chapter Five: Summary, Conclusion, and Recommendations - It is the last chapter of the project that summarises the results of the project. It determines how the system that was developed worked as per the objectives stated in linking students. It also outlines the ways to develop in the future, including the introduction of AI-based job recommendations or the development of the platform into a native iOS-Android mobile application.

1.7 DEFINITION OF KEY TERMS

In order to have a uniform interpretation of the concepts addressed during this process of documentation, the following terms are outlined within the scope of this project:

- i. Student Client (Job Poster): This is a primary user position that is the demand side of the market place. The Client will be authorized to make project listing, establish categories (e.g., Coding, Tutoring) and view applicant profiles and also make financial settlement.

- ii. Student Freelancer (Job Applicant): This is one of the primary user roles that are on the supply side of the marketplace. Following this user is the Job Feed, where he or she finds opportunities, opens applications, and a User Profile that can be considered an equivalent of verifiable digital portfolio of his or her academic and professional capabilities.
- iii. Admin: An administrative body that is highly privileged. Moderation of job content to avoid spam, user report handling, and system metrics (number of jobs, number of categories) analysis are all roles of the Admin to support the integrity of the platform.
- iv. Job Feed: An interface in the Job Management Module, which is dynamic and real-time. It shows the accessible tasks in a format of cards that are well arranged and may be filtered by deadline, the level of difficulty, and the category of project.
- v. PayStack Gateway: PayStack is a secure, third-party application programming interface (API) that is part of the system. [25].
- vi. Task Status Tracking: A graphic project management tool that takes the form of badges that are used to indicate the job lifecycle.
- vii. Responsive Web Design (RWD): An engineering practice of web development whereby the user interface (UI) of the platform is fluid.

CHAPTER 2

LITERATURE REVIEW

This chapter presents a review of related literature, including academic studies, project reports, and system analyses on the gig economy, digital labour platforms, and online freelance marketplaces. It provides insights into existing systems, highlights best practices, and identifies gaps that underscore the need for a tailored platform for student freelancers.

2.1 GENERAL INFORMATION

The "Gig Economy" has transitioned from a niche professional movement into a global economic norm. Such work is temporarily fixed, flexible, and project-based in nature, thus finding the most relevance with the present generation of college-going students, more commonly referred to as Gen-Z. Research indicates that students are increasingly interested in "micro-entrepreneurial" opportunities. These opportunities allow an individual to monetize a talent learned while in school, such as programming in Python, designing user interfaces and experiences, or tutoring school peers, without being tied to a strict 9-to-5 scheduling environment. [20].

The trio in discussion here is on three preeminent things that make a campus marketplace function: Trust, Context, and Technology. Examples of global platforms with the appropriate technology include Upwork and Freelancer, but they do not fully understand how universities function. This commonly culminates in a "Cold Start" problem, which creates difficulty for students to compete with individuals who have been doing it for years. This project addresses the issue by making the "Context" institution-specific and using Institutional Email Verification as the major way of building confidence. [21] [24]..

2.2 RELATED WORKS

The speed of research on platform, mediated labor has significantly gone up over the last ten years. Masta and Kaushiva extensively review 258 studies on platform, based work with a focus on alternative work arrangements and algorithmic supervision in the gig economy [26]. However, their synthesis hardly refers to informal and student, oriented freelance systems. Through a review of 59 academic articles, Pilatti et al. dive into the worker power relations,

collective agency, and platform autonomy mechanisms [27], whereas Alauddin et al. examine how digital labor platforms influence worker experiences and the nature of their interactions in 18 recent studies [28]. Omar and Jamil also offer a systematic rework linking digitalization, policy, and worker well-being aspects in gig labor situations [29]. Alongside these general industrial relations discussions, Fiers singles out the necessity of digital skills for freelancers to consistently be resilient when facing the changes of platform economy conditions [30], whereas Munoz et al. center their work on platform design as a factor in the perpetuation of gendered and racialized inequalities in the digital labor market [31]. Rahman and Sultanas comprehensive review of digital labor practices brings out an ethical and governance dimension to these phenomena [32]. In contrast, Brailovskaya's analysis provides a comprehensive view of how workers feel in digital gig economy [35]. There has also been a "macro" level transformation of the "labour" market. For example, Javokhir et al. looked at how the restructuring of the gig economy affects national markets using a variety of data [36]. Farah Diba et al. studied "the platforms' effect" on the ability to be flexible and the challenges of being employed in new gig segments [37], while Mako et al. addressed "the regulatory challenges with Uber" in Hungary, based on a partial study of that case [38]. Similarly, Farah Diba et al. expand on findings and theorise about well-being and the exploitation of employees due to the platformisation of work [39], while Schor documents the ways in which "platform capitalism" transformed institutions of the "sharing economy" and workforce norms [40]. Lehdonvirta puts the view of the global digital labour platforms into the context of a value chain and the wider institutional framework within which they exist [41]. Masta and Kaushiva re-examine the discussions surrounding power and autonomy within platform work [42]. Omar and Jamil reiterate the importance of digitalisation on the interaction between worker well-being and gig labour policies [43], while Baitenizov et al. investigate the entrepreneurial "spillover effects" of freelancing [44]; Wu et al. present a collection of research aggregating the "impacts" of incentives on the behaviour of gig workers [45]. Fiers further highlights resilience through qualitative inquiry into digital freelancing [46]; and the International Labour Organisation highlights the global challenges and opportunities of gig work [47].

At the very granular level, a few investigations touch upon digital marketplaces and professional freelancing. Gussek and Wiesche study how various signals can help IT freelancers win projects on digital labor platforms by looking at the large, scale empirical data [48]. Soni et al. discuss the economic and technological impact of freelancing platforms on the gig ecosystems [49], and Kim et al. define a multi, dimensional worker success concept

scoring beyond the traditional income, based one [50]. De Los Santos et al. invent sousveillance devices that can be used to fight against the algorithmic control of gig platform workers [51], while Hsieh et al. explore the interaction of personalization and standardization for both job matching and the revenue of gig workers [52]. Slightly different topics are analyzed by Mhlmann and Zalmanson as they scrutinize algorithmic control and autonomy reduction in digital labor platforms through a systematic literature review [53], and Senarath explains software waterfall methodology for sequential SDLC delivery by way of examples from the Campus Gigs software engineering model [54]. Samuel et al. contend that a well, structured digital skills training program makes graduates more employable [55], and Rahman and Sultana present the ethical and technical issues scattered across different sectors of the digital labor [56]. Rabca et al. do a comparative study of modern web development frameworks in terms of performance and suitability in software projects [57], whereas Oesterreich et al. conduct a meta, analysis of trust determinants on digital platforms [58]. Lawson investigates digitally mediated youth self, employment in Rivers State [59], and Ohira critically reviews the impact of gig labor on product development practices [60].

Another stream of relevance to employment among youth, digital participation and the adoption of platforms is the review of literature regarding digital adoption behaviours of the youth population in Nigeria and the associated labour market impacts, conducted by Oladimeji and Aminat [61] as well as Nwankwor et al., who found that entrepreneurship via social media can alleviate the issue of unemployment [62]. Gonçalves, Pereira and Coelho explored blockchain based mechanisms to facilitate the trustworthiness of transactions on digital platforms [63], while Chinyere et al., have shown how digital marketing can provide economic empowerment for youths in the Niger Delta [64]. Finally, Oyefesobi et al. found that there is a relationship between participation in social networking and employability creativity amongst students in Nigeria [65], wherein these students possess transferable digital skills for the purpose of freelancing. [22].

Taken together, the existing body of literature highlights the nature of the labour market, trust, governance of digital platforms, design, employability of youth, and digital entrepreneurship. However, the majority of the literature focuses on macro-policies and algorithmic control rather than on student-specific gig marketplaces that incorporate possibilities for academic scheduling, early-career skill development, escrowed payment, and mediation in hiring, which form the basis of the Campus Gigs design. [14] [15]

2.3 SUMMARY OF THE REVIEW

The following table provides an exhaustive review of 40 related works published within the last decade (2016–2026), categorized by their contribution to the field.

S/N	Title of Paper	Author(s)	Year	Summary	Limitations
1	Work in the Platform Economy: A Systematic Literature Review	R. Masta & P. Kaushiva	2024	Reviews 258 studies on alternative work arrangements within gig/platform economy.	May over-represent formal research, under-represents informal worker voices.
2	Systematic Literature Review on Gig Economy: Power Dynamics, Worker Autonomy & Social Networks	Pilatti, Gustavo R., et al	2024	Synthesizes 59 articles on gig work power imbalance & collective agency.	Focuses on algorithmic management rather than student-specific platforms.
3	The Influence of Digital Platforms on Gig Workers: A Systematic Review	A. Alauddin, Farah D., et al	2024	Reviews 18 articles (2019–2023) on how digital platforms shape gig work.	Geographic/industry scope limited; mostly English research.
4	A Systematic Literature Review of the Gig Economy: Worker Experiences & Policy Implications	N. Omar & R. Jamil	2025	Synthesizes 31 studies on worker well-being, policy, and digitalization influences.	Emphasizes well-being; less on platform design.
5	Resilience in the Gig Economy: Digital Skills in Online Freelancing	F. Fiers	2024	Qualitative interviews show digital skills help freelancers navigate changing platforms.	Uses small interview sample; U.S.-centric.
6	Platformization of Inequality: Gender & Race in Digital Labor Platforms	I. Munoz et al.	2023	Shows how platform design can embed social biases affecting freelancers.	Does not directly address platform economic metrics.
7	Digital Labor: Challenges, Ethical Insights, & Implications	A. Rahman & S. Sultana	2025	Comprehensive overview of digital labor trends and ethical concerns.	Broad scope; not specific to gig marketplaces.

S/N	Title of Paper	Author(s)	Year	Summary	Limitations
8	Using Computational Grounded Theory to Understand Tutors' Gig Experiences	L. Alqazlan et al.	2022	Applies computational theory to tutors' forum discussions on gig work.	Still in early phase; limited formal findings.
9	Online Freelancing on Digital Labor Platforms: A Scoping Review	P. Kim, E. Cheon & S. Sawyer	2023	Scoping review of online freelance work and design opportunities.	Not formally peer-reviewed; limited indexing.
10	Digital Labor Gig Economy: Worker Perspectives Review	Valentina Brailovskaya	2023	Reviews literature on digital gig work and platform worker experiences.	Includes gray literature and reports.
11	The Rise of Gig Economies: How Freelancing Is Reshaping Labor Markets	Javokhir M. et al.	2025	Mixed-methods analysis of gig economy impacts on labor markets.	Primarily macro-data focus, less on individual platform UI/UX.
12	The Influence of Digital Platforms on Gig Workers	Farah Diba et al.	2024	Reviews flexibility vs challenges in digital platforms for workers.	Broad region/industry variance; not student-platform focused.
13	Emerging Platform Work in the Context of the Regulatory Loophole (The Uber Fiasco in Hungary)	C. Mako et al.	2021	Literature on platform work regulation challenges, with Uber case study.	Focuses on transport platforms, not online freelancing.
14	The influence of digital platforms on gig workers: A systematic literature review	Farah Diba et al.	2025	Shows exploitative effects of platforms on worker well-being.	Limited demographic detail on highest-skilled freelancers.
15	After the Gig: How the Sharing Economy Got Hijacked	J. B. Schor	2020	Book on gig economy's evolution and institutional captures.	Not a peer-reviewed academic article.

S/N	Title of Paper	Author(s)	Year	Summary	Limitations
16	Platformization and Online Labor Markets	Vili Lehdonvirta.	2019	Research overview of global gig economy trends and indicators.	Focuses on indices, not worker experience data.
17	Work in the platform economy: a systematic literature review	Raavi Masta & Pavni Kaushiva	2024	Summarizes power, autonomy, and tech influences on gig workers.	Less explanation of platform design solutions.
18	A Systematic Literature Review of the Gig Economy: Insights into Worker Experiences, Policy Implications, and the Impact of Digitalization	Norhayati Omar & Rossilah Jamil	2025	Identifies insecurity & digitalization issues in gig jobs.	Limited by inclusion of only English-language studies, potentially excluding relevant research in other languages and regions.
19	The evolution of freelance as a new innovation business mode in the modern economy	Daniyar Baitenizov et al.	2025	Explores freelancing's role in entrepreneurship and innovation.	Theoretical focus; limited empirical backup.
20	Gig work and gig workers: An integrative review and agenda for future research.	Wu et al.	2024	Reviews external incentives motivating gig workers.	Focus limited to incentives, not systemic challenges.
21	Resilience in the gig economy: digital skills in online freelancing	Floor Fiers	2024	Digital skills help freelancers adapt & innovate on platforms.	Qualitative; small geographic sample.
22	Exploring the gig economy: Challenges and opportunities	International Labour Organization	2025	Synthesis on gig work's risks and opportunities globally.	High-level; lacks deep specialization analysis.
23	IT Professionals in the Gig Economy: The Success of IT Freelancers on Digital Labor Platforms	Lisa Gussek & Manuel Wiesche	2023	Investigates how signaling theory explains IT freelancers' success on digital labour	Focused on IT category only; not broad across gig types.

S/N	Title of Paper	Author(s)	Year	Summary	Limitations
				platforms using a large dataset of 7,166 freelancers.	
24	Freelancing Platforms: Revolutionizing the Gig Economy	V. K. Soni, E. Sinha, Y. Shinde & H. Jain	2025	Examines freelancing platforms' economic and technological impact worldwide.	Conference proceedings; not peer-reviewed journal.
25	Decoding the Meaning of Success on Digital Labor Platforms:	P. Kim, C. Asante-Agyei, I. Munoz, M. Dunn & S. Sawyer	2024	Explores how gig workers define "success" beyond income measures using longitudinal data.	Early work; limited published citation count.
26	Designing Sousveillance Tools for Gig Workers	M. De Los Santos, K. Do, M. Muller & S. Savage	2024	Discusses design of surveillance-monitoring tools to empower gig workers against algorithmic pressures.	Focused on surveillance/technology ethics, not platform economics.
27	A Little Too Personal: Effects of Standardization versus Personalization on Job Acquisition, Work Completion, and Revenue for Online Freelancers	J. Hsieh, Y. Hong, G. Burtch & H. Zhu	2022	Examines effects of platform standardization and personalization on job acquisition & earnings.	Pre-2023; not large-scale platform study.
28	Algorithmic Control and Gig Worker Autonomy: A Systematic Review	J. Möhlmann & L. Zalmanson	2021	Reviews how platform algorithms affect control and gig worker autonomy across digital labour platforms.	Focused primarily on algorithmic theory over design.
29	Preliminary Investigation on the Waterfall Methodolgy	Udesh Senarath	2021	Reviews the phases of the waterfall methodology. Discusses its	High risk and uncertainty due to integration at the end ; bottlenecks are identified late

S/N	Title of Paper	Author(s)	Year	Summary	Limitations
				rigidity and ease of management	
30	Impact of Targeted Digital Skills Training on Graduate Employability through 21st-Century Digital Competence	Samuel, Akorede & Karimu	2025	Shows structured digital skills modules improve graduates' job readiness in Ogun State.	Regional; does not examine online gig platforms.
31	Digital Labor: Challenges, Ethical Insights, and Implications	Rahman & Sultana	2025	Broad mapping of issues from crowdwork in AI training contexts.	Wide scope across sectors; not strictly gig platforms.
32	Importance of Choosing the Right Web Development Framework	Rabca et al.	2023	Computational Examines popular web framework such as React and Angular comparing their architectures, programming approaches and suitability for various project types .	Generalized framework comparison; lacks deeper technical insights or recommendations for specific scenarios
33	What Drives Individuals' Trusting Intention in Digital Platforms? An Exploratory Meta-Analysis	Oesterreich et al.	2025	Meta-analysis of trust determinants (usefulness, reputation, structural assurance) for digital platforms.	Academic meta-study; less focus on student context.
34	Internet-Mediated Marketing and Self-Employment of Youths in Rivers State	Lawson	2021	Examines how internet marketing improves youth self-employment in Rivers State context.	Older side of window; self-employment not strictly digital gigs.

S/N	Title of Paper	Author(s)	Year	Summary	Limitations
35	Literature Review of the Gig Economy from a Product Development Perspective:	Susumu Ohira	2021	Explores gig work's effect on product innovation and management.	Focus on product context vs labor issues.
36	The Growing Digital Footprint of Nigerian Youth: Implications for Engagement & Employment	Oladimeji & Aminat	2024	Reviews youth digital adoption patterns, internet use, job seeking and socio-economic impacts.	Broader digital engagement; not direct gig platform analysis.
37	Social Media Entrepreneurship and Youth Unemployment Reduction in Nigeria	Nwankwor et al.	2024	Empirical study showing that social media content creation generates jobs and reduces youth unemployment.	Focused on social media, not traditional gig marketplace.
38	User Reputation on E-Commerce: Blockchain-Based Approaches	Gonçalves, Pereira & Coelho	2022	Reviews blockchain reputation systems to secure user trust in online platforms.	E-commerce focused; may generalize to gig marketplaces.
39	Online Marketing Services and Socio-Economic Emancipation of Youths in Niger Delta Region	Chinyere, Ojunim & Orji	2024	Finds online marketing services correlate with increased self-employment and financial stability for regional youth.	Regional focus; not nationwide data.
40	Social Networking and Students' Academic Creativity: Implication on Youth Employability	Oyefesobi, Adetunji & Oladunjoye	2020	Links academic social networking use to creativity and employability outcomes among Nigerian students.	Not gig platform-specific but relevant to digital engagement.

Table 2.1 Summary of the Review

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

In this chapter, the system analysis and design for Campus Gigs will be presented. This will include the methodology used, system specifications, data models, functional system requirements, system architecture, and design diagrams. A detailed blueprint for developing the freelance platform for students while ensuring utmost security and efficiency, along with the justifications for the design selections made during the development phase, will be presented.

3.1 METHODOLOGY

The methodology stipulates the procedures, methods, and development methodology that is to be employed in the software solution implementation. The software development methodology is a complementary mechanism used in the process of selecting different processes that need to be undertaken in the development of the solution beginning with the requirements gathering up to implementation, testing, deployment, and maintenance of the solution. In Campus Gigs, a site that provides students with the opportunity to promote their freelancing services, it had to determine the process distinctly, as the system was of multifunctional character, as it worked at the tiers of a freelancer, client, and administrator. The most appropriate model to be used in this project is the Waterfall Software Development Life Cycle (SDLC) model. The quality of

the system, maintenance, and integration of a system, as noted by the software engineering, determines the methodology applied to the system [66]. Waterfall offers a progressive and sequential model that can be defined as having specific steps such as requirements analysis, design, implementation, testing, deployment and maintenance [67]. Its linear model is appropriate in the case of projects whose requirements are constant and their deliveries are predictable [68], in which the completion of one stage leads to the start of the other.

3.1.1 Justification for Methodology

The Waterfall approach was deemed appropriate and justifiable based on a number of considerations that relate to the scope and academic-oriented nature of the Campus Gigs project. Initially, the system was understood to require minimal changes in the system requirements. The functionalities of the system, including the posting of jobs, applications for those jobs, messaging, payment systems, and overall management, do not require an exploratory approach. They were readily understood, and a focus on documentation is a key concern, which is also an important academic requirement[19].

Thirdly, Waterfall offers predictability in terms of schedules. In academic projects, a fixed semester period is common, and project deliverables must be realized in stages of completion in line with academic examination periods. Iterative development processes may involve rework loops, which might to an extent negate the academic project schedule. Fourthly, as Waterfall development allows for multiple user support, this calls for proper design and data modeling to occur from early stages of project development, thereby emphasizing the advantage of Waterfall's design before code approach. [71]. Moreover, Waterfall is helpful in academic software development projects in that it allows for ease of explanation of design decisions by the students. [72]. Lastly, Waterfall reduces miscommunication in small development teams by eliminating dynamic requirement changes, which can otherwise increase risk.

3.2 SYSTEM ANALYSIS

System analysis is a process of examining what is and what can be done in operations to discover inefficiencies, set up user requisites, and justify system intervention . Campus Gigs is a great example for the analysis of how students are currently conducting freelance transactions outside the formal platforms and what kind of inefficiencies these systems have. The goal is to figure out the flow of information, labor, payments, and communication without a digital intermediary and to check what interventions the system proposes.

System analysis also helps to identify the relevant actors (students, clients, administrators), assess stakeholder expectations, record issues, and derive technical requirements. The new systems feasibility, therefore, becomes quantifiable and justifiable through this process, thus laying the groundwork for the next phases in the design cycle.

Research studies on informal gig work focus on information fragmentation, reputation asymmetry, and payment insecurity as main barriers for freelancers , especially young people who are just entering the labor market .

3.2.1 Analysis of Existing System

Because student freelancers are not connected by any type of organized workflow, they must utilize informal and decentralized methods (i.e., social media, a network of fellow student freelancers, and their respective social networks) to locate potential freelance opportunities. Opportunities are most often discovered at a time when the student is actively engaged with other members of their fraternity/sorority or club network. Employment terms (e.g., type of work, timeframe for completion, number of revisions, deliverables, and payment amount) are negotiated manually through private messages with little, to no, formalized proof of previous work for potential future engagements. Students and clients also have no formalized systems for establishing client trustworthiness or verifying student skills/competency. As a result, the current structure relies on personal trust and reciprocity rather than formal contracts, which creates issues related to inefficiencies and disputes. Since communication between student freelancers and clients occurs in a decentralized manner, there is no way to collect data on these interactions to analyze trends, and there is no central authority that can mediate disputes. Additionally, students' academic obligations create a barrier to balancing academic and freelance work, and there is no formalized system to assist students in managing their workload.

3.2.2 Limitations of Existing System.

The existing informal system has many structural flaws:

- i. **Fragmented Access to Jobs:** The freelance prospects exist in informal avenues and this makes it inefficient to match skilled students with the corresponding jobs.
- ii. **Payment Vulnerability:** In the absence of escrow or intermediate payment validation, the students will be exposed to non-payment or late compensation.
- iii. **Gaps in Dispute and Accountability:** No arbitration mechanism is established, no agreements are documented, and hence, it becomes hard to resolve conflicts.
- iv. **Reputational Invisibility:** The students are not able to use their portfolio evidence or rating to gain market credibility.

- v. Inefficient Scheduling: The system fails to bring in academic limitations like exams, coursework due dates or semester. [79].
- vi. Unequal Visibility: Chances are skewed towards socially connected students, which lacks inclusiveness.
- vii. Absence of Administrative Governance: There is no supervision to govern unfair practices and minimum standards of engagement

All these restrictions are a valid reason to create a student-specific freelance marketplace that would automatize such procedures.

3.2.3 Justification of New System

Campus Gigs aims to address the problems caused by the absence of a structure in terms of informal freelance offers by creating a centralized online platform, where students could find and request freelance employment. Campus Gigs also offers such opportunities under one roof and provides all students with an equal opportunity of accessing the freelance market without considering their social network. Portfolios of students will be rated, and this will enhance their skill of signaling their marketability and reducing the information asymmetry that exists between clients and freelancers.

The integration of payments will eradicate the financial risk that both customers and freelancers will have and this will improve professional confidence between them. Admin tools will give the clients an opportunity to settle their disputes and will enable Campus Gigs to keep an eye on whether its guidelines and policies are adhered to and will eventually make its clients feel confident with the Campus Gigs platform. Scheduling tools enable the students to manage their workload (freelance) with their courses to avoid burnouts or time conflicts. To conclude, Campus Gigs facilitates increased transparency, operational rationality, equity, and marketability ability among students with more access to qualified freelancers by clients.

3.2.4 Description of the New System

Campus Gigs is a web-based freelance job marketplace that is run by students and enables customers to post jobs that require a deadline and a pay rate. Learners can find tasks and apply to and submit tasks that they are interested in. After selecting a student to do a task, the student and the client exchange messages to make clarifications and agree on the final deliverables. [85]. Campus Gigs integrates payment processing (through Paystack) to offer secure payment options to clients and the history of payment transactions that they can use in the future. An

administration panel offers control over the use of platforms and helps to solve conflicts and ensure that users comply with the platform policies. Users maintain profiles containing skills, completed work records, ratings, and reviews, allowing clients to evaluate freelancers based on competence rather than personal networks. The system employs status-tracking for jobs (posted, applied, accepted, completed) and supports post-job evaluations.

3.3 SYSTEM SPECIFICATION

System specification determines the technical requirements, limitations, and working conditions necessary for the successful implementation of the Campus Gigs platform. As our solution is a web, based student freelancing marketplace, the specifications are centred around performance, scalability, interoperability, and maintainability. Having a well, articulated specification stage guarantees that the development will be in line with the stakeholder expectations and that the system components will be compatible throughout the deployment.

3.3.1 Hardware specification

Being a web, based app, campus gigs does not significantly require hardware for development and testing. We used personal computers that had:

- i. Processor: a modern CPU with a minimum clock speed of 2.4 GHz (e.g., Intel Core i5 or equivalent) to handle compilation, debugging, and testing tasks efficiently.
- ii. Ram: 4 GB RAM (8 GB recommended for smooth multi-process development).
- iii. Storage: The system needs at least 20 GB of free storage capacity to store project files combined with dependencies.
- iv. Internet connection: stable internet connectivity for package installation, repository management, and api integration.

3.3.2 Software specification

We chose a software technology stack for the development of the Campus Gigs platform that would be easy to integrate, maintain, and scale culturally. Our software requirements include the operating system, programming languages, development tools, frameworks, third, party services, and supporting utilities required during implementation and operation. The specifications also reflect user and performance demands identified during our analysis phase. The development environment has been implemented using the code editor, termed Visual Studio Code (VS Code), owing to its extensibility, intellisense features, and scripting capabilities, as well as debugging features and version control using tools based on Git. Further, we have used Git and GitHub tools for versioning and documentation throughout the development process. For the browser-based debugging and layout tests, we have used Chrome Dev Tools due to its capacity to profile network calls and optimize the loading process. Its architecture is grounded on the standard multi-tier model with the following basic technologies:

- i. **Frontend Technologies:** The presentation layer of the application was implemented using HTML, CSS and JavaScript. These languages help us to create responsive user interfaces, content organization and asynchronous communication between the browser and the server with the AJAX technology.
- ii. **Backend Server Technologies:** The technology used to implement the backend server is the Node.js technology. Express.js has been chosen to be the main one that will be used to implement business logic, routing, and middleware services. The reason why node.js was selected is that of the non-blocking I/O nature of Node.js which is capable of managing real-time user interactions concurrently. REST based architecture is also compatible in node.js. Also, Node.js ecosystem has mature libraries, package support, and good community documentation which allows it to develop and maintain quicker.
- iii. **Database Technology:** The system has incorporated the use of MongoDB, which is a NoSQL data database. There is no strong schema enforcement in MongoDB, and the diverse data structures including job posts, bids, notifications, messages, and user profiles can be stored in the database. These operations are ever beneficial in the case of marketplace systems where data structures can evolve at any time.
- iv. **Payment Gateway Integration:** An integration of the Paystack API was used to support online payment between students, acting as clients, and freelancers. Paystack supports payments made through cards, bank transfers, and other local means, hence

ideal for use in Nigeria. Additionally, Paystack uses web hooks that enable verification of transactions and updating of the ledger.

- v. Testing and Communication Tools: For testing the backend API endpoints, Postman has been used.

3.4 DESIGN OF PROPOSED SYSTEM

This section explains how our structure is based on the needs and requirements of the user, software engineering practices, and the way the marketplace operates. System design helps us to effectively visualize how the data will flow, the logic behind it, and the way the user interacts with the systems.

3.4.1 Data Model

The Data Model defines what information entities interact with each other, and how that information is structured on the Campus Gigs platform. The Data Model is the basis for implementing the Campus Gigs platform, and captures the entire Freelance Workflow (Job Posting - Submission, Payment, and Feedback) logically. Traceability of all User Actions, Platform Transactions, and Administrative Oversight is necessary for maintaining and scaling Marketplace Systems

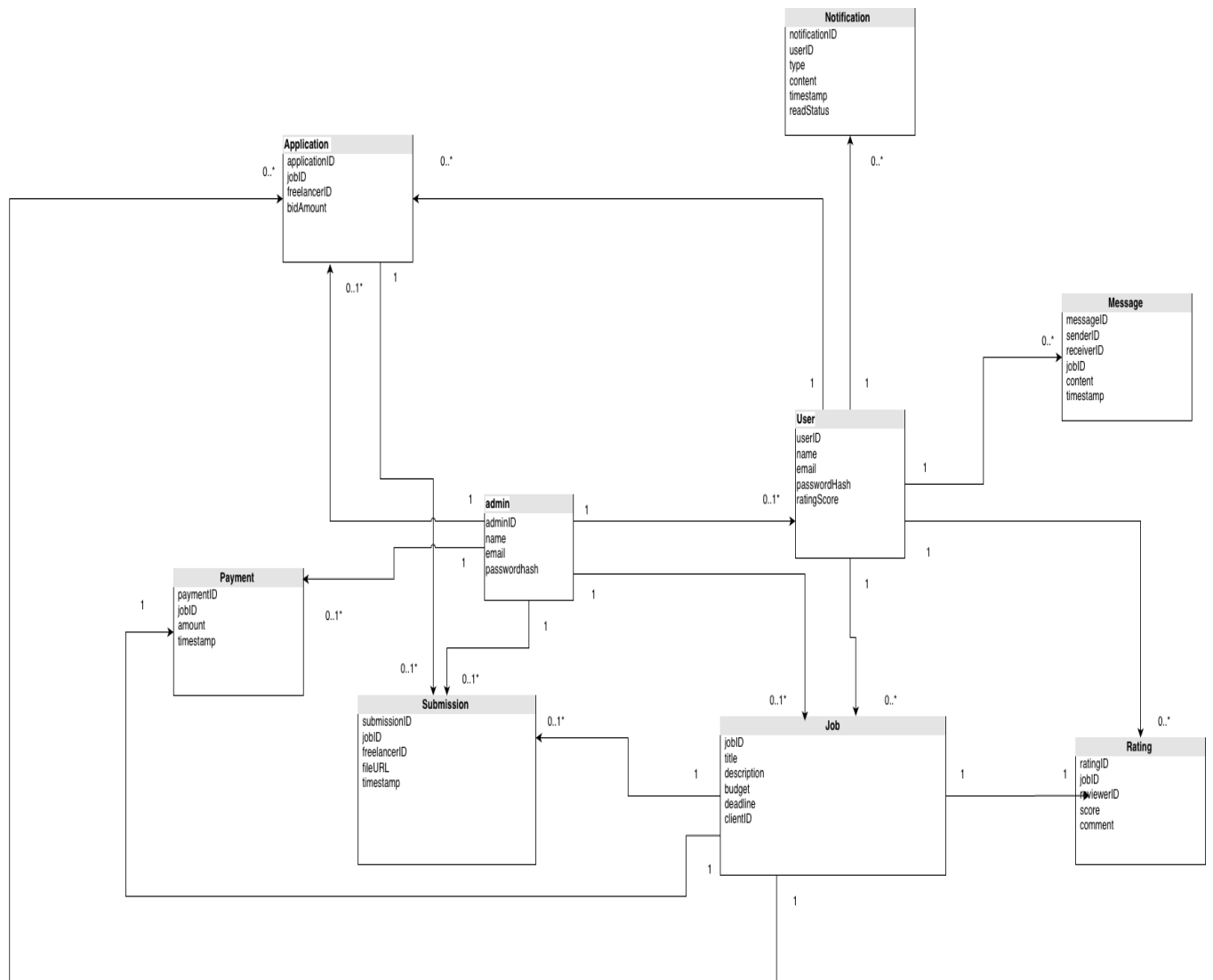


Figure 3.1: Entity-Relationship Diagram (ERD)

In the case of Campus Gigs, the data model that was completed is an eight-core data model based on the functional and non-functional requirements that were developed during the analysis phase. These entities include:

User: A person who collaborates with the site. There are three possible roles that a user can play in our system: that of a client, freelancer and administrator. Clients create jobs, freelancers create and perform jobs, whereas the administrators moderate and resolve disputes. The fields we incorporate include ratingScore to aid the reputation and accountability processes.

Job: Refers to the activities of freelance jobs, which are generated by customers. Attributes of jobs include budget, deadlines and status flags, which enable us to check the lifecycle progress (open - assigned - submitted - completed). All jobs should be linked to a single client but possibly have numerous freelancers during the application process.

Application: The application section encompasses applications of freelancers interested in taking up a job posted. The applications enable us to support the selection, negotiation and bidding. The client may reject or accept an application, and only an accepted application will proceed to the execution of tasks.

Submission - The submissions can be seen as deliverables by freelancers once an application has been made. This body is in charge of making sure that the site records evidence of job completion, uploaded files, timestamps, and revising cycles in the event that a client seeks amendments prior to making the final decision.

Payment: The financial transactions are related to jobs. Payments are indicative of escrow behaviour, where some money is held up when a job is delegated and discharged when the submission is approved. This provides equity, responsibility and protection of compensation of the two parties.

Rating: It is a review and feedback after the completion of a job. This body aids the reputation system and provides incentive on good work based on points. The ratings are associated with the reviewer and the job in question.

Message: This is a form of communication between users in the process of negotiation, clarifying or in the execution. Messaging helps to coordinate in real time and make sure that any dispute or misunderstanding is possible to be corrected because of the conversation logs stored.

Notification - Refers to automatically generated notifications that the system makes users aware of updates to their jobs, changes in application status, release of applications, release of payments, or feedback. Notifications enhance the responsiveness of the platform and minimise delays in communication.

The following entities represent the lifecycle of freelancing:

User - Job posting - Application- Assignment - Submission - Payment- Rating.

3.4.2 FUNCTIONAL REQUIREMENTS

Functional requirements define what the Campus Gigs system should be doing to meet the requirements of the users, the stakeholders and the operations. In our project, the functional requirements were obtained in Chapter 3.2 by the system analysis and proved later in Chapter 1.5, as shown in the diagram below with the use case modelling. The demands are in direct relation to the transactional process of a student freelancing marketplace, which enables clients, freelancers, and administrators to post, apply and execute tasks, and ensure compliance and moderation.

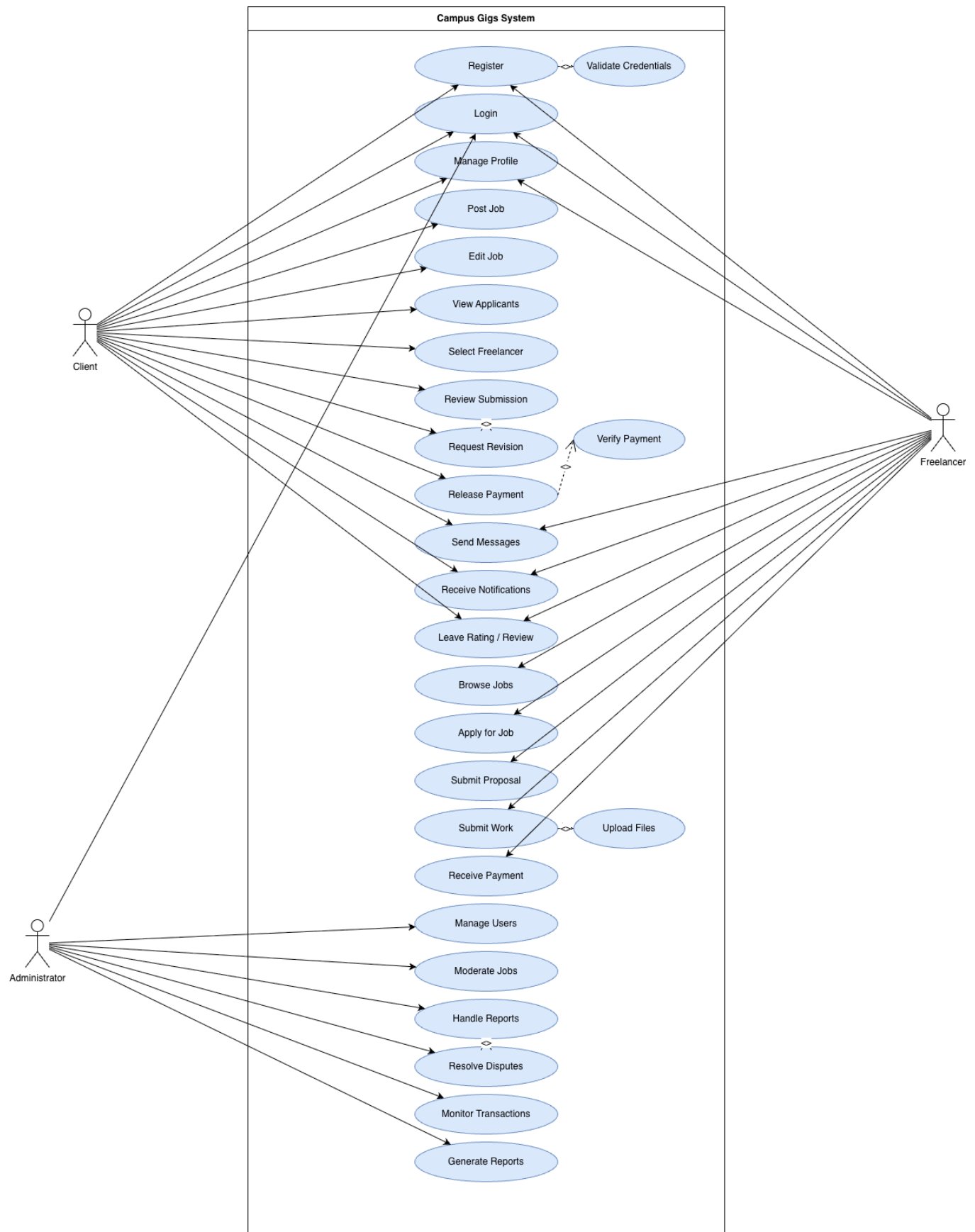


Figure 3.2: Use Case Diagram

Actors

Campus Gigs involved three main stakeholders, which we determined:

Job Poster (Client): Refers to those students who need to fulfil academic or practical assignments. Job posters generate job advertisements, screening of candidates, authorisation of outputs, and payments.

Freelancer (Student Applicant): Students who provide services on a paid basis. The freelancers find jobs, bid to work on tasks, interact with their clients, deliver their work and get graded in terms of the work.

Administrator (Admin) : Designates system administration charged with the responsibility of platform management. The administrator is in charge of user accounts, content moderation, dispute resolution and system or operational reporting.

These actors represent a common role division of the marketplace used in gig economy sites and enable us to decouple issues of job creation, job execution, and operational administration.

Functional Requirements

1. The users must be capable of registering and signing up to get an account using valid credentials.
2. Users ought to have the capability of accessing secured features on the platform through a secure means of accessing the platform.
3. The users are supposed to access their profile and update them in order to have the correct personal and academic details.
4. Job posters (student clients) must have the ability to post jobs giving specifics of job like description of task, budget, and deadline.
5. Job posters must be able to modify posted jobs in order to update the requirements or modify job information where needed.
6. Each job posted should have the capability of viewing the applicants through job posters.
7. Job posters must be in a position to filter the applicants of a particular job and decide on the freelancers.

8. Job posters ought to have the capability of updating job status during the freelancing lifecycle (e.g., assigned, submitted, completed).
9. Freelancers (student applicants) would be created to see job feeds having available open jobs.
10. Freelancers ought to have a chance of rendering interest and proposal details on the job.
11. Users are expected to be able to negotiate, clarify tasks or request revisions by chatting or messaging each other.
12. The users ought to be able to get the notification about their updates like applications, approvals, submissions, or payments.
13. The freelance workers ought to be capable of delivering the work outputs to the client to be reviewed and accepted.
14. Job posters must have the capability of accepting work submitted through evaluation of task deliverables.
15. Job posters must be in a position to process a payment via the inbuilt payment gateway upon the approval of the tasks.
16. Accountability and reputation should be encouraged on the platform by allowing the user to leave ratings or reviews.
17. The administrators ought to be in a position to administer users, jobs and reports to maintain system integrity, moderation and dispute resolution.

3.4.3 SYSTEM ARCHITECTURE

The system architecture for Campus Gigs adopts a distributed web-based model designed to support scalability, modularity, real-time interaction, and secure payment processing. The platform is deployed within a cloud environment and is accessible through a standard web interface via HTTPS connections. The architecture is partitioned into the presentation layer, application layer, data layer, and external service integrations, ensuring functional separation and maintainability.

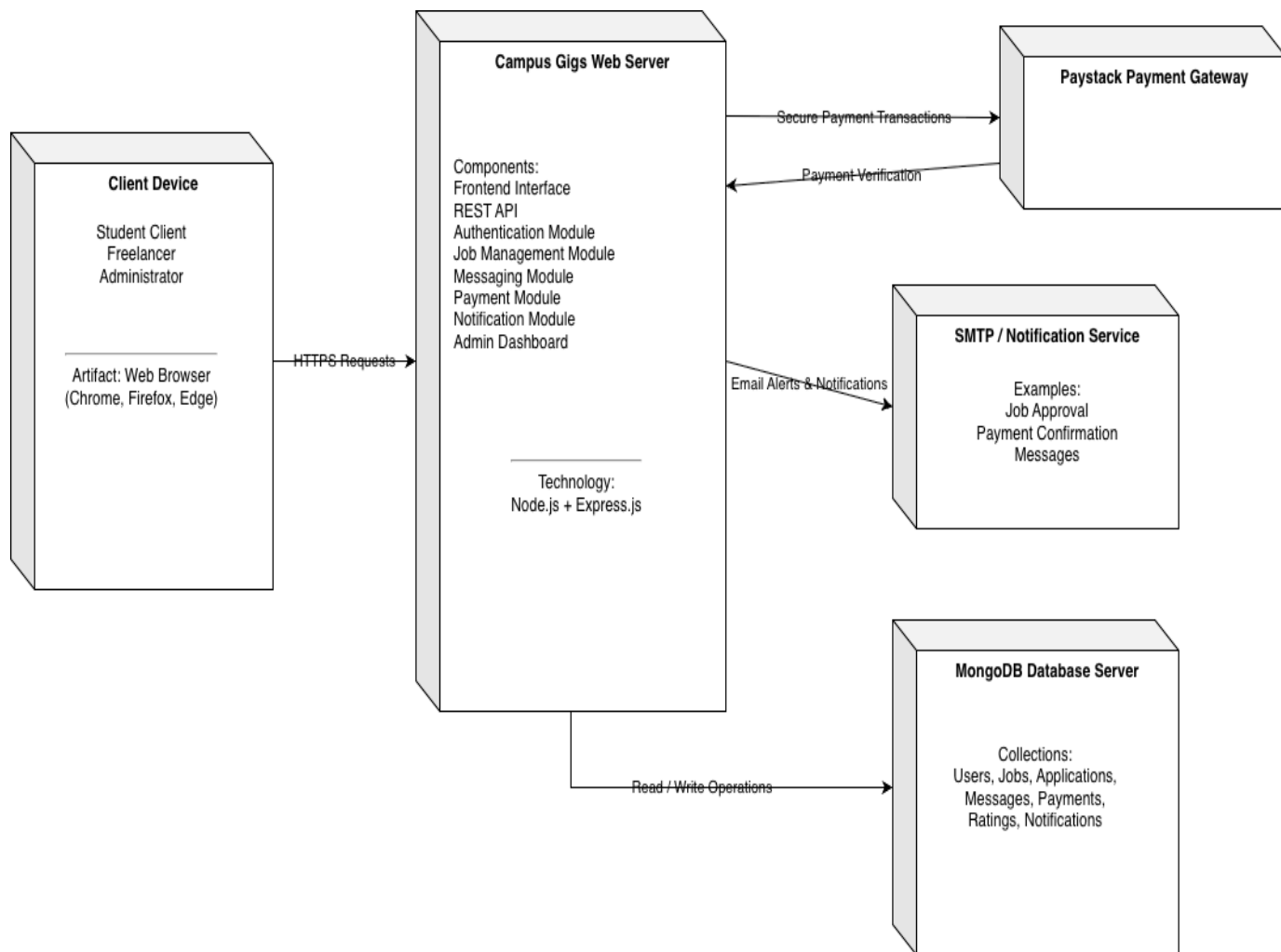


Figure 3.3: Deployment Diagram

The presentation layer includes the web client, which serves as the interface for the students playing the role of freelancers or clients. Users can create accounts, post jobs, apply for jobs, submit deliverables, send payments, receive notifications, etc. All interactions between the client and the server are done via RESTful API endpoints over secure communication channels. The application layer uses Node.js with the Express framework, which manages business logic, requests, authentications, jobs, messaging, payments, etc. The backend also includes the integration of the notification module, which supports sending emails for job status, user verification, task assignments, etc. The data layer includes the MongoDB database, which stores user accounts, job postings, applications, messages, payments, ratings, etc. A document-oriented schema was chosen, which supports flexible data modelling. This type of schema can handle dynamic data models, which is appropriate for this problem domain.

External services are part of the architecture, which allows critical functionalities to be achieved. This includes an online payment gateway that will facilitate an escrow mechanism between clients and freelancers, and an SMTP service that will facilitate email communication. Additionally, an administrative panel will be included in the architecture, which will allow moderation of users, jobs, reports, and disputes. Overall, it will facilitate the end-to-end lifecycle of freelancing activities in a university setting, from job posting and application to submission, approval, and payment, in a way that is reliable and extensible.

3.4.4 Software Structure

The Campus Gigs software architecture is a modular domain-based model which is used to map platform objects to base application classes. Business logic, attributes and relational operations used to support the freelancing lifecycle are encapsulated in each major class. A class-based architecture enables us to decouple interface issues and business issues, and enables us to keep adding features in the future without breaking the rules of maintainability.

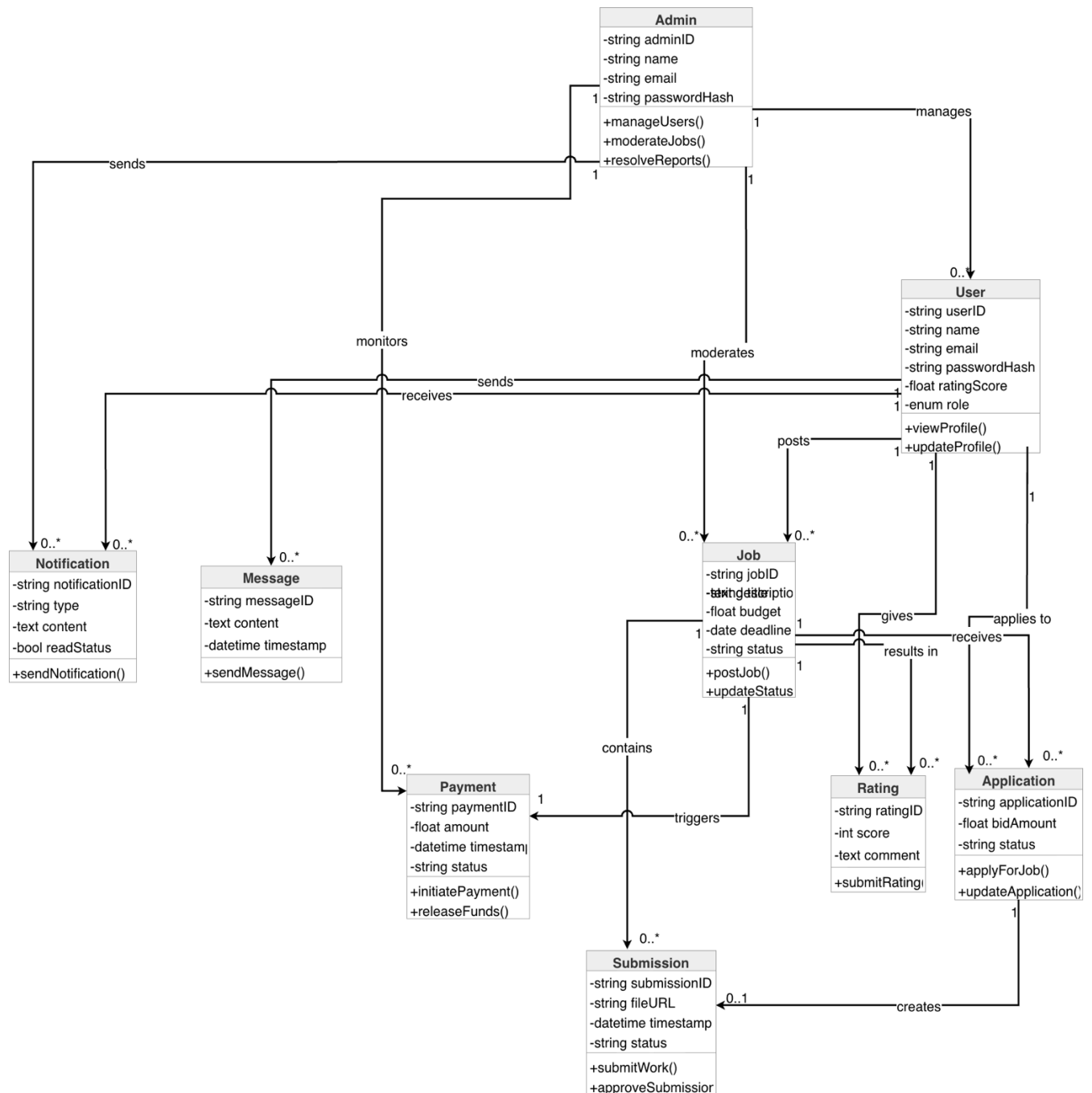


Figure 3.4: Class Diagram

The domain-level model, which forms the basis of the Campus Gigs software structure, is formalised by the class diagram. Every type of class is associated with a functional organisation that advances the operations of the freelancing marketplace. Classes that are based on the ERD are the User, Job, Application, Submission, Payment, Rating, Message, and Notification classes, which are discrete objects that are handled by application services in the architecture.

The definitions given to methods like `postJob`, `apply for Job`, `submit work`, `release funds` and `submit rating` are one of the ways in which business logic is encapsulated internally to the Backend API and Application Layer components. As an example, the `Payment` class talks to the `Payment Service` to handle the escrow-like transaction, and `Submission` talks to `Job` and `Notification` to handle the revision or approval steps.

This structural decomposition is consistent with the component diagram presented above, that is, service modules `messaging service`, `job service`, `application service`, and `payment service` communicate with these domain classes to realize functional requirements. The class diagram thus constitutes the conceptual base connecting the data model of the platform with its modules of operation services and finally with its interfaces of the UI.

3.4.5 Activity Diagram

The activity diagram is used to model the dynamic workflow of the freelancing lifecycle at Campus Gigs. It shows operational states and changes that are made between job posting and payment release, and rating submission. The activity diagrams are used to validate the sequences of interactions between the actors and to ensure consistency among business rules, application, lifecycle events.

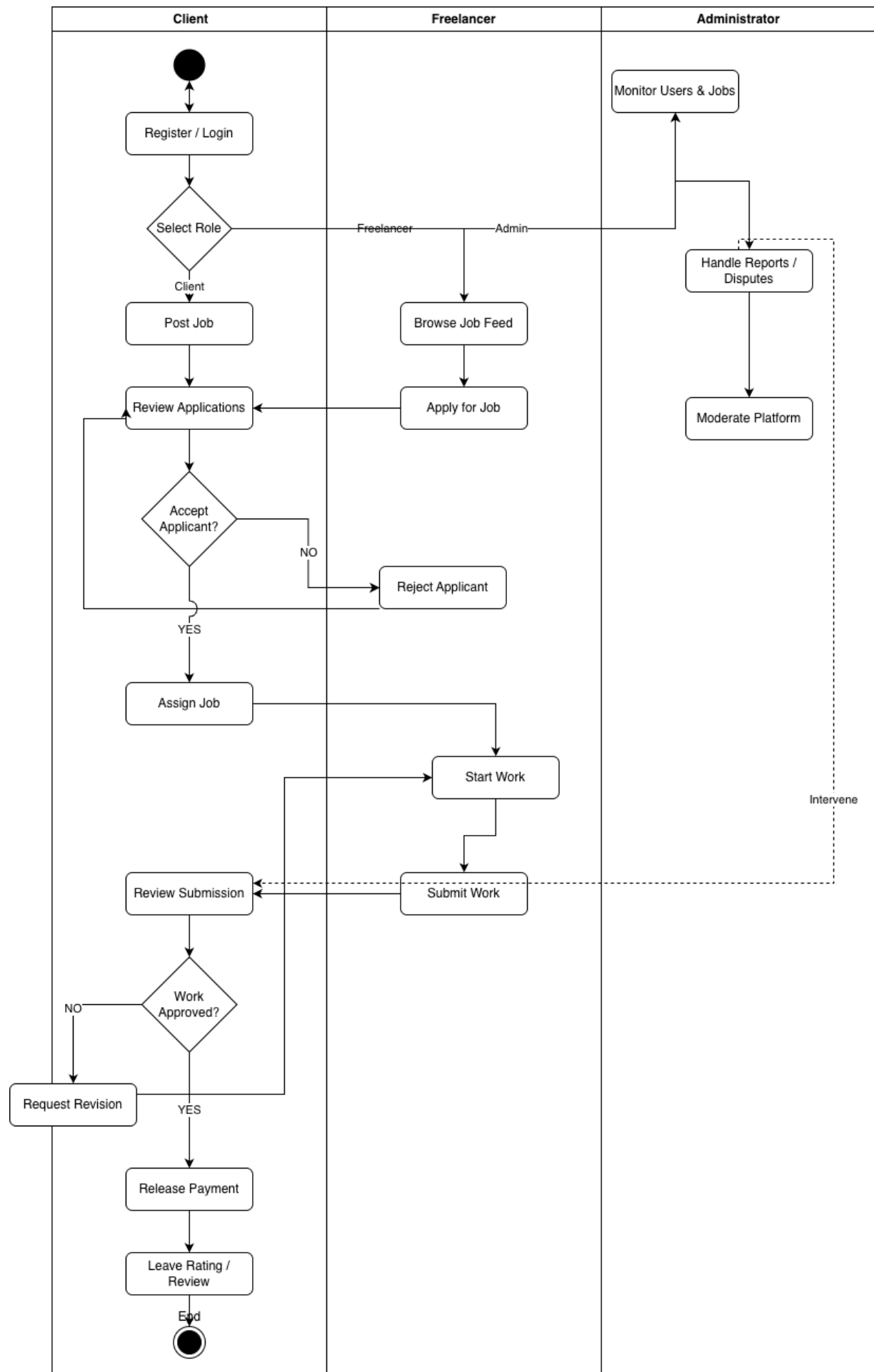


Figure 3.5: Activity Diagram

The activity diagram represents the changing state of the core marketplace workflow and decision paths. The paperwork starts with the posting and advertising of the job to the freelancers and thereafter the applications may be handed in by the freelancers. After the approval of an application, escrow semantics are used to reserve payment to provide integrity of transactions. The freelancer does the given task and sends deliverables to the client. In case of approval, the job is transferred to the completed state and the money is given out. In case the client demands changes, the process of work starts over again, and it continues to make corrections until the approval.

This is the workflow which corresponds to the lifecycle of freelancing, which is supported by the software structure and corresponds to the objects in the class diagram, namely Job, Application, Submission, and Payment. It also strengthens the messaging, notification and revision of requirements to ensure that the design of the system promotes collaboration, accountability and safe payment processing.

CHAPTER FOUR

IMPLEMENTATION AND DISCUSSION

This chapter presents the implementation and discussion of the Campus Gigs system. The chapter explains how the system was developed in accordance with the design specifications and requirements discussed in the previous chapter. It also discusses the technologies, development environment, frontend interfaces, backend database structure, and testing procedures used during the implementation process. The implementation phase transformed the system's conceptual and architectural design into a fully functional web-based freelance marketplace platform specifically for students of Nile University of Nigeria. The implementation process involved integrating frontend and backend technologies to create a secure, responsive, and scalable platform capable of handling core system functionalities, including user registration, authentication, job posting, job applications, messaging, notifications, task submission, payment processing, and administrative moderation. This chapter also evaluates the system's performance and usability through functional and non-functional testing conducted during development.

4.1 SYSTEM REQUIREMENT FOR DEVELOPMENT

The development of the Campus Gigs platform required a combination of hardware and software resources to ensure smooth implementation, testing, deployment, and maintenance of the system. These requirements were carefully selected to support the technologies used in developing the application while ensuring efficiency and scalability throughout the software development lifecycle.

The hardware requirements for developing the system included a modern computer system with sufficient processing power to support programming, database management, testing, and debugging. A computer system with at least an Intel Core i5 processor or its equivalent was used during development because of its ability to efficiently handle multiple development processes simultaneously. The system also required at least 4 GB of RAM, though 8 GB was preferred for better performance when running development servers, databases, browsers, and testing tools concurrently. Storage space was also important during implementation because the system required enough space to accommodate source code files, project dependencies, database packages, frameworks, and other development utilities. At least 20 GB of available

storage space was sufficient for these operations. In addition, a stable internet connection was necessary during development to facilitate package installation, cloud services integration, repository synchronisation, and third-party API implementation, such as Paystack integration.

The software requirements consisted of the tools, frameworks, programming languages, and utilities used in implementing the platform. The development environment was implemented on Windows and macOS operating systems because of their compatibility with the selected technologies. The frontend aspect of the application was developed using HTML5, CSS3, and JavaScript. These technologies were chosen because they provide the flexibility required to create responsive and interactive web interfaces that are compatible with different devices and browsers. The backend implementation of the Campus Gigs was developed using modern server-side technologies that support scalability, efficiency, and secure data management. The backend architecture of the system was implemented using Node.js together with the Express.js framework. Node.js was selected because of its event-driven and non-blocking input/output architecture, which enables the server to efficiently handle multiple user requests simultaneously without significant performance reduction. This feature made Node.js highly suitable for the development of a dynamic freelance marketplace platform where multiple users may interact with the system concurrently through messaging, job applications, notifications, and payment transactions.

Express.js was integrated as the primary backend framework because it simplified server-side development by providing routing mechanisms, middleware support, request handling, and API management functionalities. The framework enabled the implementation of RESTful APIs that facilitated communication between the frontend interfaces and the backend database. Express.js also improved the organization and maintainability of backend code through its modular structure and middleware architecture.

MongoDB was used as the database management system for the application. The database was selected because of its flexibility, scalability, and document-oriented data storage structure. Unlike traditional relational databases, MongoDB stores information in JSON-like documents, making it suitable for handling dynamic data structures associated with users, jobs, applications, messages, payments, notifications, and reviews within the platform. The flexibility of MongoDB also simplified data modelling and allowed the system to scale efficiently as more features and users are introduced. The connection between the backend server and the MongoDB database was implemented using Mongoose, which served as the

Object Data Modelling (ODM) library for the application. Mongoose simplified schema creation, validation, querying, and relationship management within the database. It also improved data consistency by enforcing predefined structures for collections such as users, jobs, applications, and transactions.

Several supporting development tools and technologies were utilised during the backend implementation process. Visual Studio Code was used as the primary integrated development environment because of its debugging capabilities, extension support, and code management features. Git and GitHub were used for version control and collaborative project management, enabling efficient tracking of code modifications throughout development. Postman was used extensively for testing API endpoints and verifying communication between the frontend and backend systems. Google Chrome Developer Tools were also utilised for debugging and monitoring application performance during implementation.

To support secure online transactions within the platform, the Paystack API was integrated into the backend payment module. The integration enabled payment initialisation, transaction verification, and payment status management between clients and freelancers. Security mechanisms such as encrypted password storage, token-based authentication, protected routes, and server-side validation were also implemented within the backend architecture to improve system security and protect user information from unauthorised access.

Overall, the selected backend technologies provided a stable, scalable, and efficient environment for implementing the core functionalities of the Campus Gigs platform while ensuring maintainability, performance optimisation, and secure communication between system components.

4.2 SYSTEM MENUS IMPLEMENTATION

The frontend implementation of Campus Gigs focused on creating a user-friendly, responsive, and visually appealing interface that would enable users to interact effectively with the system. The system interfaces were designed according to responsive web design principles to ensure compatibility across desktops, tablets, and mobile devices. The frontend implementation also emphasized usability, accessibility, simplicity, and smooth navigation to improve user experience.

The landing page served as the entry point into the system. The interface was designed to provide users with a clear understanding of the purpose and objectives of the platform. The landing page contained a navigation bar, hero section, service highlights, call-to-action buttons, and footer sections containing additional platform information. The hero section displayed a collaborative student-themed design that aligned with the project's concept of connecting student freelancers and clients. The landing page also provided direct access to registration and login functionalities.

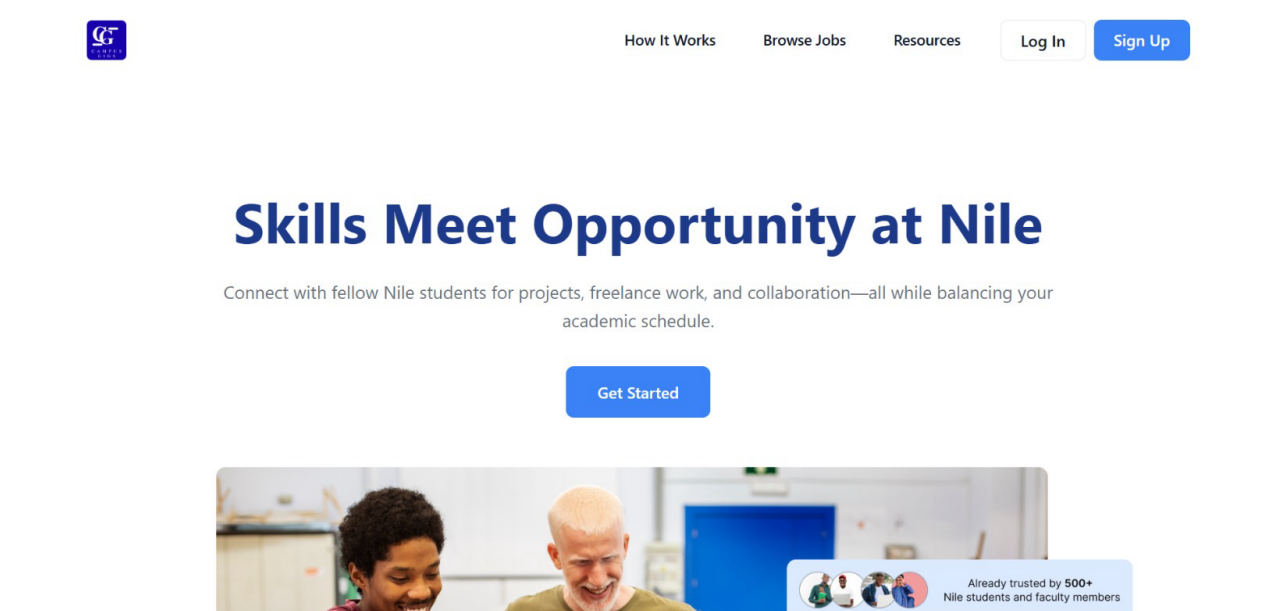


Figure 4.1 Landing Page

The registration interface was implemented to allow students to create accounts securely within the platform. The registration form collected important user information such as full name, email address, password, department, and role selection. Input validation mechanisms were implemented to ensure that users entered valid and complete information before account creation. Password security measures were also implemented during registration to ensure that user credentials remained protected within the system database.

Create an account

Exclusive marketplace for students to offer skills and find campus opportunities

Student Email

Please fill out this field.

Student ID Number

Password

Confirm Password

[Sign Up](#)

Already have an account? [Log in](#)

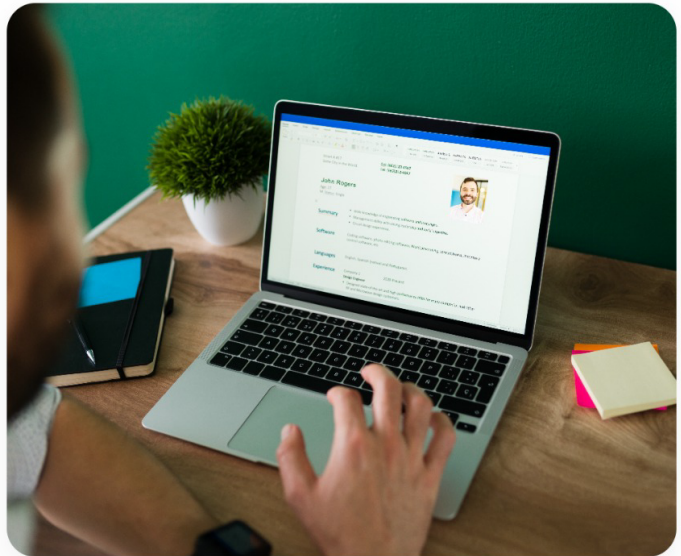


Figure 4.2 Registration Interface

The login interface enabled registered users to securely access the platform using their credentials. Authentication logic was implemented on the backend server to verify user information before granting access to secured areas of the application. The login process also incorporated encrypted password comparison and session management mechanisms to improve platform security. Appropriate error handling was implemented to notify users whenever invalid credentials were entered.



Log in

Log in to find flexible gigs that fit your class schedule and start your career before graduation.

Student Email

Password

[Forgot password?](#)

Log in

Don't have an account? [Sign Up](#)

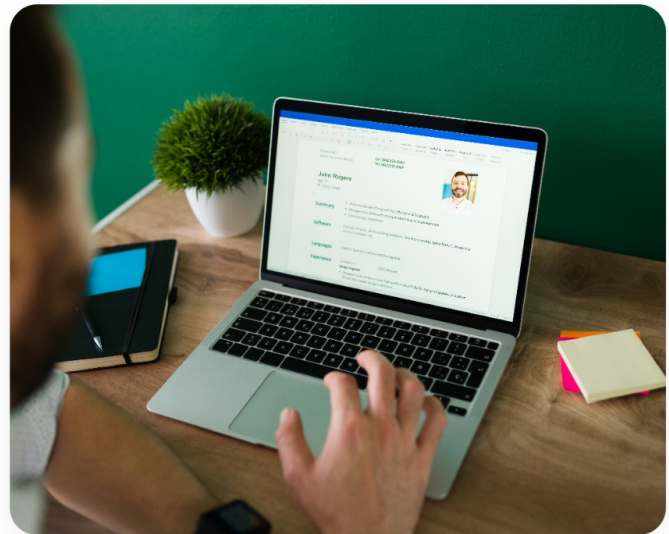


Figure 4.3 Login Interface

The user dashboard interface provided users with personalised access to platform activities based on their assigned roles. Freelancers were presented with dashboards containing available jobs, applications, notifications, earnings summaries, and messaging shortcuts. Clients had access to dashboards that displayed posted jobs, applicant information, project management features, and payment management tools. The administrator dashboard provided administrative functionalities such as user moderation, job monitoring, report handling, and system analytics.

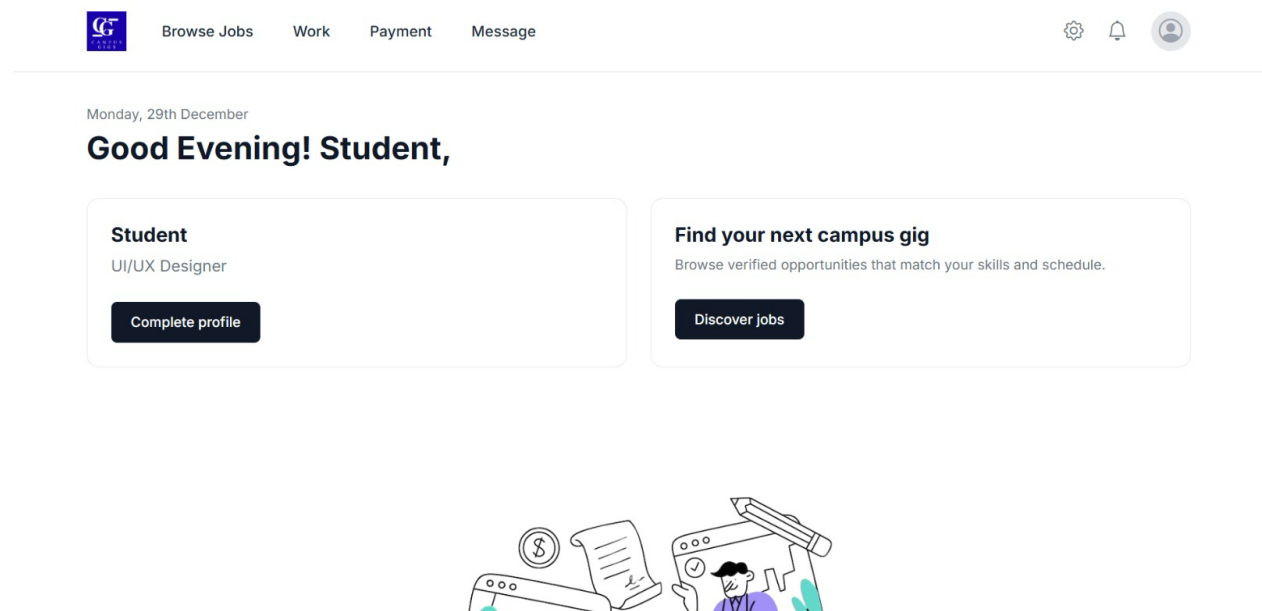


Figure 4.4 User dashboard

The job posting interface allowed student clients to publish freelance opportunities on the platform. The interface contained fields for entering job title, description, category, budget, deadline, and required skills. After submission, the system validated and stored the job information within the database before displaying it in the public job feed accessible to freelancers.

Post a New Job

Provide details about the gig to attract the right talent.

Job Title *

e.g. Need a UI/UX Designer for a Mobile App

Category * **Project Type ***

Select a category ▼ Fixed Price ▼

Job Description *

Describe the project, scope of work, and your expectations...

Be as detailed as possible to get the best proposals.

Required Skills

e.g. Figma, HTML, CSS (Separate with commas)

Figure 4.5 Job Postion Interface

The job feed interface was implemented to provide freelancers with access to available freelance opportunities. The interface dynamically retrieved jobs from the backend database and displayed them in a structured card-based layout. Search and filtering functionalities were implemented to allow freelancers to search for jobs based on category, deadline, and budget preferences. The application functionality was integrated directly into the job feed to simplify the application process.

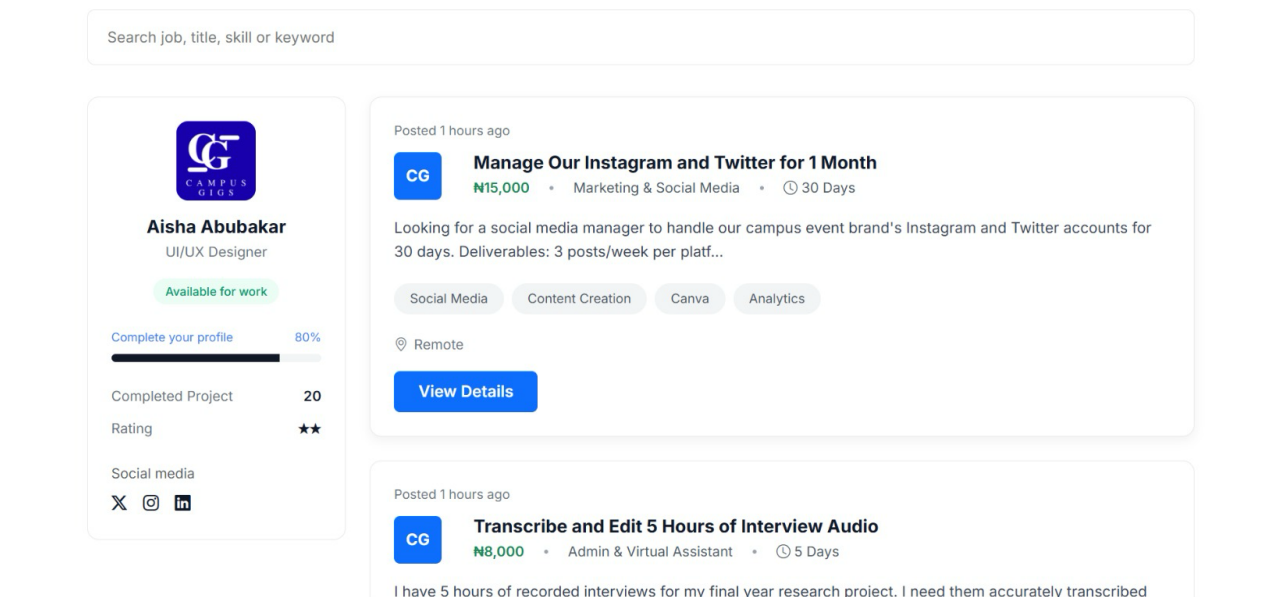


Figure 4.6 Job feed Interface

The application interface allowed freelancers to submit proposals for jobs they were interested in. Freelancers could provide proposal descriptions, bid amounts, and estimated completion durations when applying for jobs. The system stored submitted applications within the database and made them accessible to clients for review and selection.

The screenshot shows a web form titled "Your Proposal". It contains several sections: a "Cover Letter" section with a text area and a character count (0 / 2000); a "Your Bid (₹)" section with a text input field showing "e.g. 25000" and a note "Client's budget shown above for reference."; a "Delivery Time (days)" section with a text input field showing "e.g. 7" and a note "How many days do you need?"; and a "Resume / Portfolio (optional)" section with a "Choose File" button and a "No file chosen" status. A yellow warning message states: "File upload coming soon — configure Cloudinary first. For now, include portfolio links in your cover letter." At the bottom is a blue "Submit Proposal" button.

Your Proposal

Cover Letter *

Introduce yourself and explain why you're the right person for this job. Mention relevant experience, your approach, and any questions you have. Minimum 50 characters.

Min 50 characters 0 / 2000

Your Bid (₹) *

e.g. 25000

Client's budget shown above for reference.

Delivery Time (days) *

e.g. 7

How many days do you need?

Resume / Portfolio (optional)

Choose File No file chosen

⚠ File upload coming soon — configure Cloudinary first. For now, include portfolio links in your cover letter.

Submit Proposal

Figure 4.7 Application Interface

The messaging interface was implemented to facilitate communication between freelancers and clients throughout the freelancing lifecycle. The messaging module supported real-time communication, message storage, and conversation history management. This functionality improved collaboration between users and reduced misunderstandings during project execution.

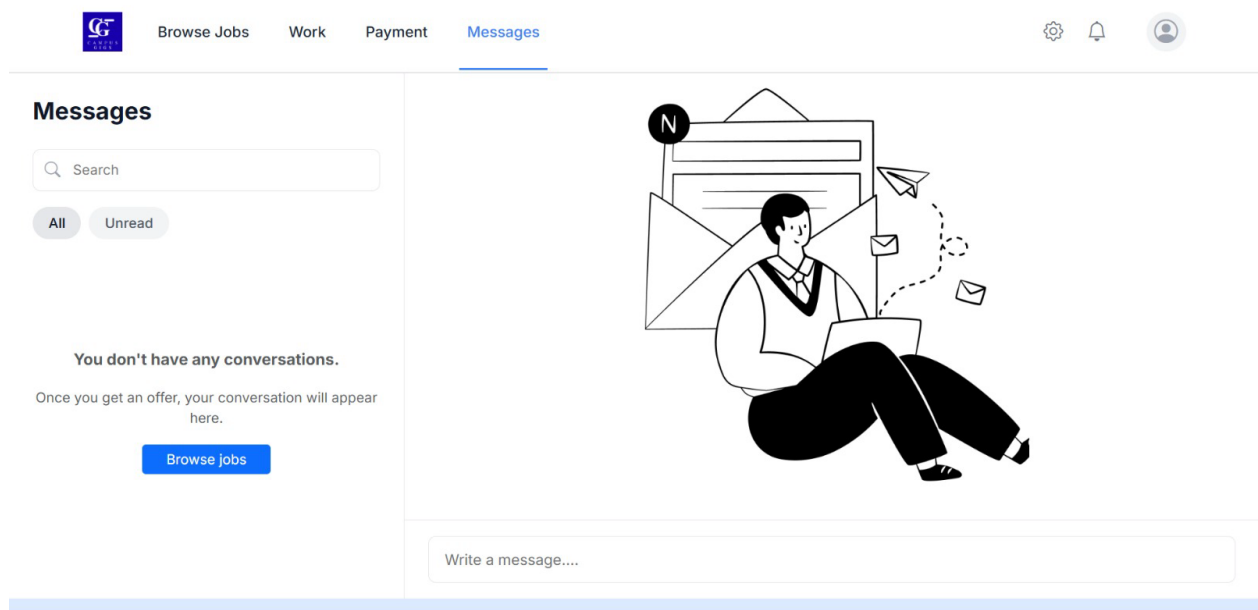


Figure 4.8 Messaging interface

The payment interface integrated the Paystack payment gateway to support secure online financial transactions. Payment processing functionalities included payment initialisation, transaction verification, and payment history management. The implementation ensured that freelancers received compensation after successful completion and approval of assigned tasks.

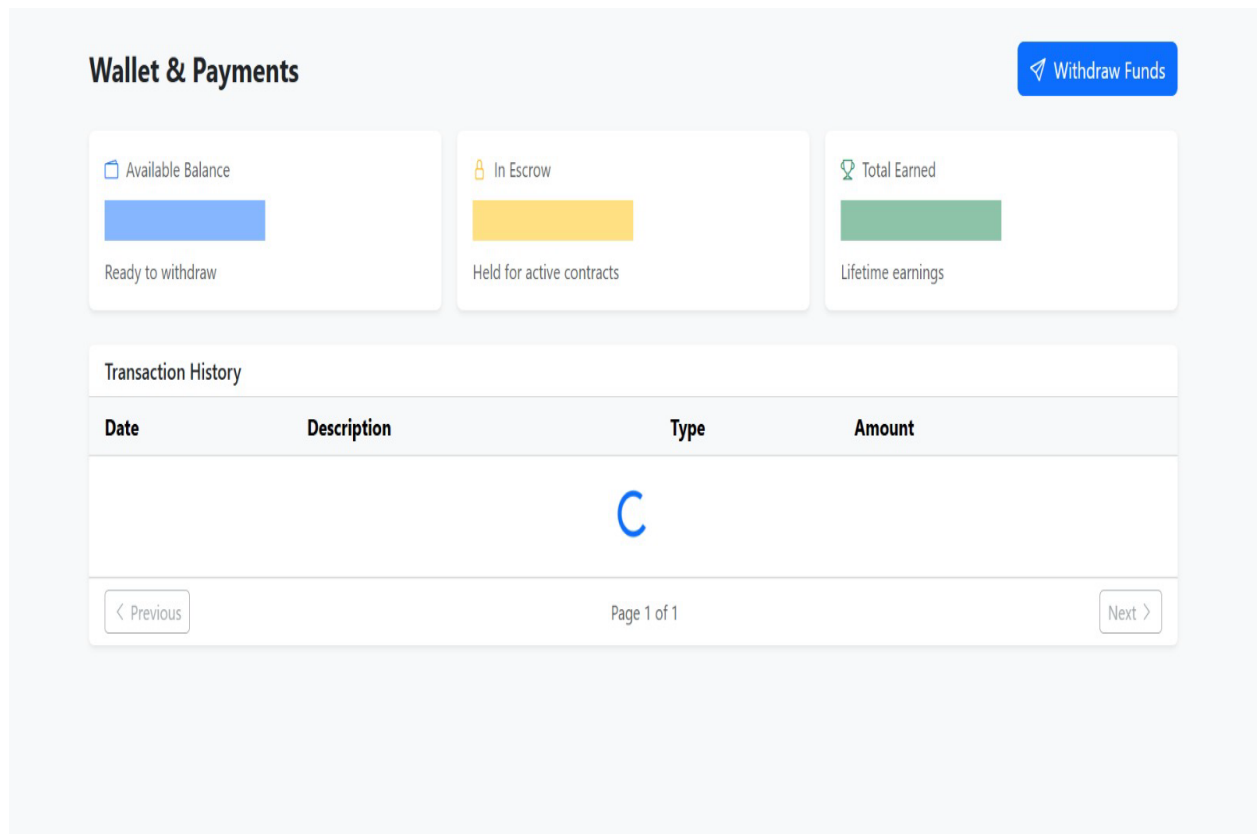


Figure 4.10 Payment interface

The notification interface was implemented to provide users with real-time updates concerning job applications, approvals, submissions, payments, and messages. Notifications improved platform responsiveness and ensured that users remained informed about important system activities.

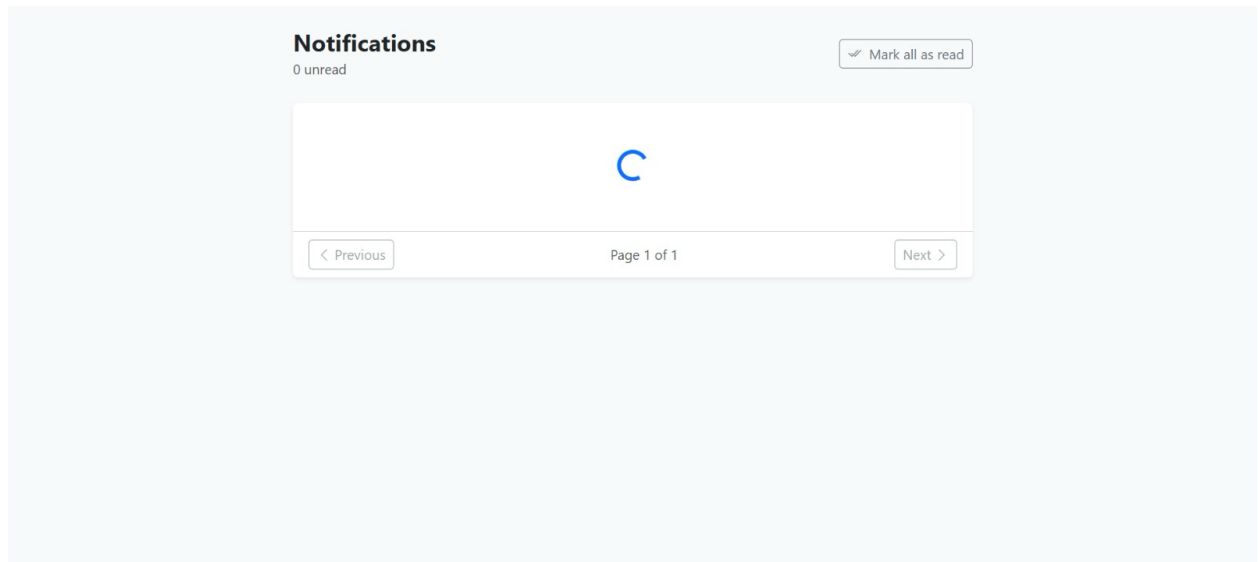


Figure 4.11 Notification Interface

The Administrator Dashboard is a centralised control panel that allows system administrators to monitor and manage all activities within the Campus Gigs platform. It provides tools for managing users, moderating job postings, handling reports and disputes, monitoring transactions, and generating system reports, ensuring the platform operates securely and efficiently.

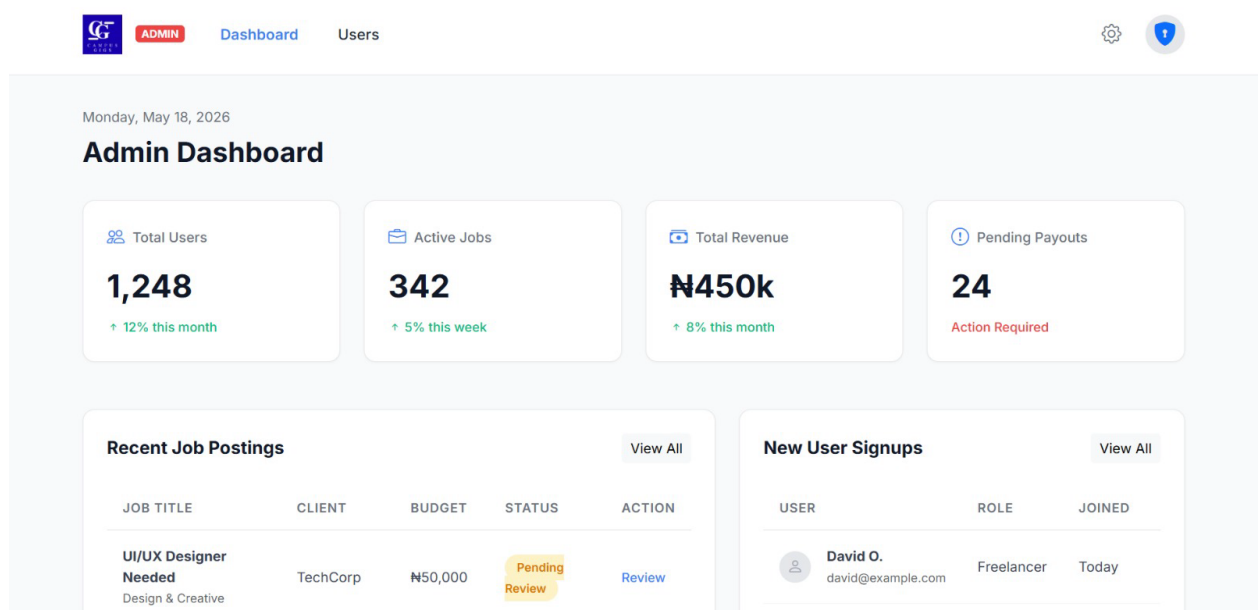


Figure 4.12 Adminstrator Dashboard

4.3 DATABASE IMPLEMENTATION

The backend implementation of Campus Gigs focused on creating a scalable and flexible database architecture capable of supporting the operations of a student freelance marketplace. MongoDB was selected as the database management system for its ability to efficiently store unstructured and dynamic data using document-oriented collections.

The database implementation involved creating multiple collections to represent the system's core entities. These collections included users, jobs, applications, submissions, messages, payments, ratings, and notifications. Each collection was designed to support specific

functionalities within the freelancing lifecycle while maintaining efficient relationships between entities through unique identifiers.

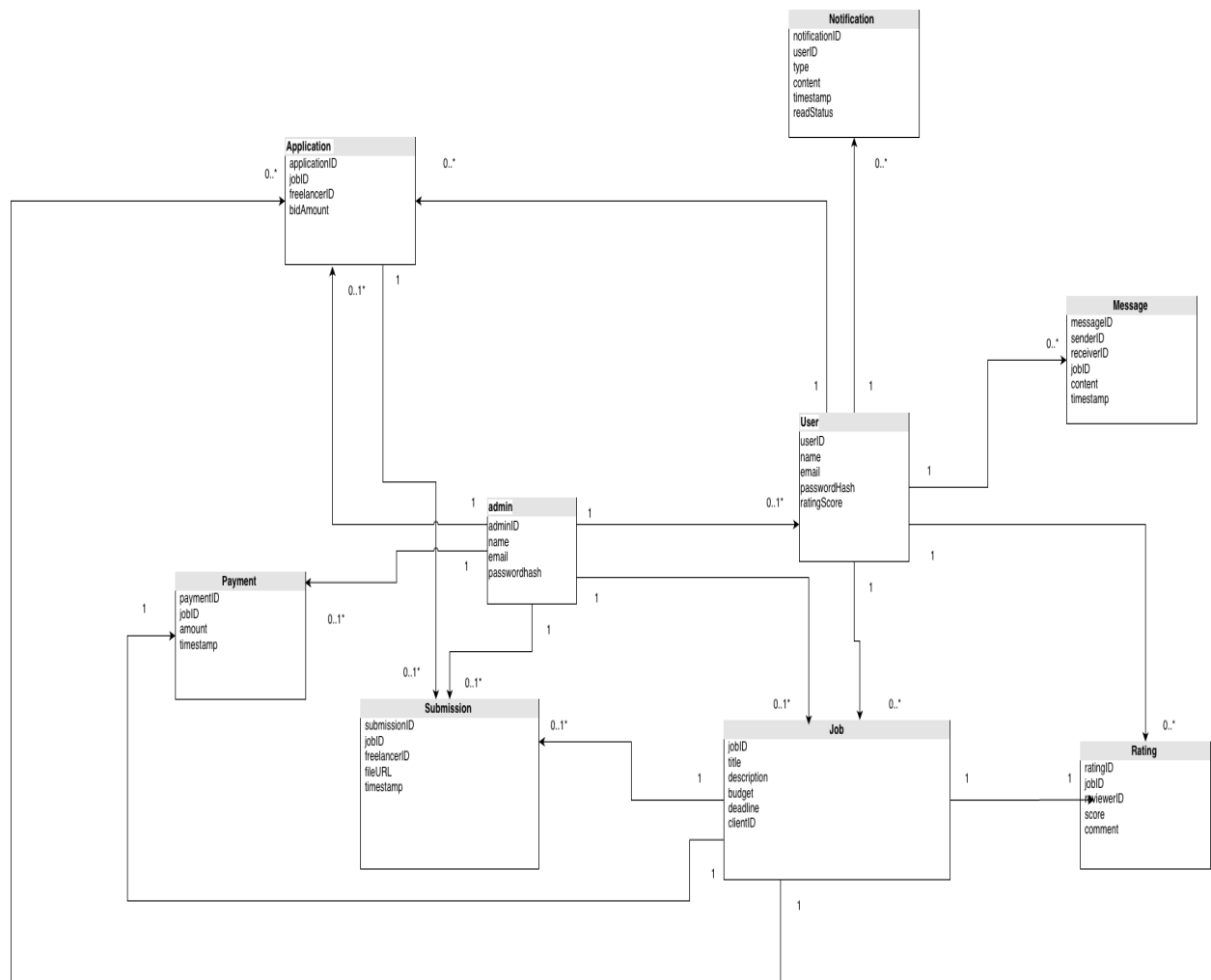


Figure 4.12: Entity-Relationship Diagram (ERD)

The user collection stored information relating to registered users within the system. The stored data included user names, email addresses, encrypted passwords, departments, skills, user roles, profile images, and rating scores. Password encryption mechanisms were implemented to improve security and protect user information from unauthorized access.

The job collection managed all freelance opportunities created by student clients. The collection stored information such as job title, description, budget, deadline, category, job

status, and client identifiers. Relationships between jobs and users were maintained through unique user IDs to ensure traceability and accountability.

The application collection stored freelancer proposals submitted for available jobs. This collection contained information such as application identifiers, freelancer identifiers, proposal descriptions, bid amounts, and application statuses. Clients could retrieve and review applications through this collection before selecting freelancers for projects.

The messaging collection managed communication records exchanged between users. The collection stored sender identifiers, receiver identifiers, message content, and timestamps. This implementation supported the real-time communication functionalities integrated into the platform.

The payment collection stored transaction information relating to financial activities conducted within the platform. The stored information included payment identifiers, user identifiers, transaction references, payment amounts, and payment statuses. The Paystack API was integrated into the payment system to enable secure transaction verification and processing.

The notification collection stored alerts generated automatically by the system whenever important activities occurred. These notifications informed users about applications, approvals, submissions, messages, and payment confirmations. Notification storage improved communication efficiency and user engagement throughout the platform.

The backend server was connected to MongoDB using the Mongoose Object Data Modeling library. Mongoose simplified schema creation, data validation, query management, and relationship mapping within the application. RESTful APIs were developed using Express.js to enable secure communication between frontend interfaces and backend database resources.

4.4 SYSTEM TESTING

System testing was conducted to evaluate the reliability, performance, usability, security, and functionality of the Campus Gigs platform. Both functional and non-functional testing techniques were employed to ensure that the developed system met the objectives and requirements identified during the analysis and design phases.

Functional testing focused on verifying that each feature of the system operated according to specification requirements. The registration module was tested to confirm that new users could successfully create accounts using valid credentials. The login module was tested to ensure that only authenticated users could access secured system functionalities. The job posting module was tested to verify that clients could successfully create and publish freelance opportunities on the platform.

The application functionality was tested to confirm that freelancers could apply for available jobs successfully. The messaging module was tested to ensure that users could exchange messages without communication failures. File upload testing was also conducted on the submission module to verify that freelancers could submit completed tasks successfully. Payment processing testing was performed to validate transaction initialisation and verification through the Paystack payment gateway. Notification functionalities were tested to ensure that system alerts were delivered appropriately whenever major activities occurred within the platform. Administrative functionalities were also tested to confirm that administrators could manage users, jobs, and reports effectively.

Non-functional testing focused on evaluating system usability, performance, responsiveness, compatibility, and security. Usability testing was conducted by allowing selected students to interact with the platform and provide feedback regarding ease of navigation, interface simplicity, readability, and overall user experience. The results indicated that users found the system easy to navigate and visually understandable.

Performance testing evaluated the speed and responsiveness of the application during operation. The platform maintained acceptable loading times and handled multiple user requests efficiently under moderate usage conditions. Security testing was conducted to evaluate authentication mechanisms, password encryption, transaction protection, and unauthorised access prevention. The results confirmed that user information and financial transactions were securely protected.

Compatibility testing was also conducted across different web browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge. The system functioned properly across these browsers without significant display or functionality issues. Responsiveness testing further confirmed that the user interface adapted correctly across desktops, tablets, and mobile devices.

The testing procedures conducted during implementation demonstrated that the Campus Gigs platform successfully met its functional and non-functional requirements. The integration of secure authentication, messaging, payment processing, notifications, and administrative management functionalities contributed significantly to the efficiency and reliability of the platform. Overall, the results obtained from the testing process confirmed that the system is suitable for deployment within an academic environment and capable of supporting student freelance activities effectively.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

This chapter presents the summary, conclusion, and recommendations of the study carried out on the design and implementation of Campus Gigs. The chapter highlights the major findings and achievements of the research while evaluating the extent to which the objectives of the study were achieved. It also provides recommendations for future improvements and possible areas of expansion that can further enhance the effectiveness, scalability, and usability of the system within academic environments. The study focused on developing a web-based freelance marketplace platform designed specifically for students of Nile University of Nigeria. The system was created to bridge the gap between students seeking freelance opportunities and those requiring digital services within the academic community

5.1 SUMMARY OF THE STUDY

The rapid growth of digital technology and remote work opportunities has significantly transformed the way individuals access employment and freelance services globally. However, despite the increasing popularity of freelancing platforms, many students within academic institutions face challenges in accessing reliable and secure environments where they can offer their skills, gain practical experience, and connect with potential clients within their educational communities. This challenge formed the basis for the development of the Campus Gigs system.

The study began by identifying the limitations associated with traditional methods of student freelancing and service exchange within universities. These limitations included difficulty in connecting skilled students with potential clients, lack of centralized communication systems, insecurity in financial transactions, absence of accountability mechanisms, and limited opportunities for students to build professional portfolios and practical experience. Existing general-purpose freelancing platforms were also observed to be less tailored toward the unique needs and environments of university students. To address these problems, the study proposed the development of a web-based freelance marketplace platform specifically designed for students. The proposed system was intended to provide a secure and accessible digital environment where students could register, create profiles, post jobs, apply for freelance

opportunities, communicate with clients, submit completed tasks, receive payments, and build professional reputations through ratings and reviews.

The literature review examined various concepts related to freelancing systems, online marketplaces, web-based applications, digital collaboration platforms, and electronic payment systems. The review also explored related studies and existing platforms while identifying gaps that justified the need for a student-centered freelancing solution. Several technologies and methodologies relevant to modern software development were also analyzed during the study.

The system analysis and design phase involved identifying the functional and non-functional requirements of the platform. Various system modeling tools such as use case diagrams, activity diagrams, entity relationship diagrams, and system architecture diagrams were used to illustrate the structure and operational flow of the proposed system. The study adopted a modular and scalable architecture that separated frontend operations, backend logic, database management, and external service integrations.

The implementation phase transformed the conceptual design into a functional software application. The frontend component of the system was developed using HTML5, CSS3, and JavaScript to create a responsive and user-friendly interface. The backend component was implemented using Node.js and Express.js, while MongoDB served as the database management system. The system also integrated Paystack APIs to facilitate secure payment processing between users. Several core functionalities were successfully implemented within the platform. These functionalities included user registration and authentication, profile management, job posting, job applications, messaging, notifications, task submissions, payment processing, ratings and reviews, and administrative moderation features. The interfaces were designed to ensure simplicity, usability, responsiveness, and accessibility across different devices and browsers.

Comprehensive testing procedures were conducted to evaluate the functionality, security, performance, compatibility, and usability of the system. Functional testing confirmed that the major features operated according to the specified requirements, while non-functional testing verified the reliability, responsiveness, and security of the platform. The testing results demonstrated that the system was capable of supporting student freelance interactions effectively and securely.

Overall, the study successfully achieved its aim of designing and implementing a centralized student freelance marketplace platform capable of facilitating collaboration, skill exchange, project management, and digital entrepreneurship within an academic environment.

5.2 CONCLUSION AND RECOMMENDATIONS

The development of the Campus Gigs demonstrated the importance of technology-driven solutions in addressing challenges associated with student freelancing, collaboration, and digital service exchange within universities. The study successfully designed and implemented a web-based platform that provides students with opportunities to connect, collaborate, and engage in productive freelance activities within a secure and organized environment.

The system successfully integrated major functionalities required for a modern freelance marketplace, including secure user authentication, job posting and application management, messaging, notifications, task submissions, payment processing, and administrative moderation. These functionalities collectively contributed to the creation of a platform capable of improving communication, transparency, accountability, and efficiency among users.

The implementation of the system also demonstrated the effectiveness of modern web technologies such as Node.js, Express.js, MongoDB, HTML5, CSS3, and JavaScript in building scalable and responsive web applications. The adoption of a modular backend architecture and document-oriented database structure improved flexibility and maintainability while supporting future scalability requirements.

The study further established that a dedicated student freelance platform can contribute significantly to skill development, entrepreneurship, financial empowerment, and practical learning among students. By providing students with opportunities to apply their academic knowledge in real-world scenarios, the system promotes innovation, professional growth, and career readiness within the university community.

Despite the successful implementation and testing of the system, certain limitations were identified during the course of the study. The platform was developed primarily for web-based access and may require additional optimization for native mobile applications in the future. Furthermore, the current implementation focused mainly on core freelancing functionalities and did not extensively incorporate advanced features such as artificial intelligence recommendations, video conferencing, automated dispute resolution, or blockchain-based transaction verification.

Based on the findings and outcomes of the study, several recommendations are proposed for future improvements and enhancements. Future versions of the system should include dedicated mobile applications for Android and iOS platforms to improve accessibility and user convenience. The integration of artificial intelligence algorithms for job recommendations, freelancer matching, and personalized content delivery can also improve user experience and platform efficiency.

Additional security enhancements such as multi-factor authentication and advanced fraud detection mechanisms should be implemented to further strengthen user protection and transaction security. Future implementations may also incorporate video conferencing tools to facilitate better communication and remote collaboration between freelancers and clients.

It is also recommended that future research explore the integration of machine learning techniques for predictive analytics, reputation management, and intelligent project recommendations. Integration with institutional student databases may further improve verification processes and reduce the possibility of unauthorized access to the platform.

Furthermore, universities and educational institutions should encourage the adoption of platforms such as Campus Gigs as part of broader digital entrepreneurship and skill development initiatives. Such platforms can create opportunities for students to gain practical work experience, develop professional networks, and generate income while pursuing their academic studies.

In conclusion, the Campus Gigs system successfully achieved its intended objectives by providing a secure, responsive, and functional web-based freelance marketplace tailored specifically for students. The system addressed several challenges associated with student freelancing and demonstrated the potential of digital platforms in promoting collaboration, innovation, entrepreneurship, and professional development within academic communities. The study, therefore, contributes meaningfully to the advancement of student-centred digital solutions and provides a strong foundation for future improvements and expansions in the field of online freelance marketplace systems.

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APPENDIX I

APPENDIX II